Generative AI and Data Quality: Implications for Productivity, Labor Displacement, and Policy

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Abstract

The success of generative AI depends on high-quality, human-created data. Yet, as AI adoption grows, an increasing share of new data is being generated by AI itself. This paper examines the economic impact of this transition through a social learning model of endogenous AI adoption, where both AI and human labor can produce and their production activities generate data of potentially different qualities. A key feature is the AI-data feedback loop: data quality affects AI productivity, which influences AI adoption and, consequently, the composition (AI- v.s. human-generated) and quality of future data.

Drawing on recent experimental findings in computer-science research on generative AI, the calibrated model predicts hump-shaped labor market dynamics following a cost reduction of AI. Labor displacement can be significant in the short run but partially reverses over time as low-quality AI-generated data degrades future datasets. The equilibrium is inefficient because individuals do not internalize the impact of their actions on overall data quality. Taxing AI adoption or limiting AI's access to data may enhance overall welfare.

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1 Introduction

Modern artificial intelligence (AI) has achieved remarkable success by learning from vast collections of human-created data. Large language models derive their understanding of language structure and meaning from billions of pages of human-written texts—from books and articles to online content. AI image generators have learned to recognize objects, textures, and scenes by training on enormous databases of photographs. This foundation of high-quality human-created data has been crucial for AI to generate relevant and accurate outputs for real-world applications (Liu et al., 2024).

This situation is changing. The rapid adoption of generative AI has led to an unprecedented surge in AI-generated content. The European Union Law Enforcement Agency projects that AI-generated content could constitute up to 90% of online content in the long term.[1] Rio-Chanona et al. (2023) shows that the introduction of large language models like ChatGPT has reduced human-generated content on platforms like Stack Overflow, while Brooks et al. (2024) documents an increasing presence of AI-generated content on Wikipedia, often of inferior quality.

This trend suggests that future AI will likely be trained on a mixture of AI-created and human-created content—if human-created content remains available at all.[2] This has raised concerns among both AI researchers and the broader community. Shumailov et al. (2024) and many other computer science studies (summarized in Table 1[3]) have documented "model collapse"—a phenomenon where the quality of AI-generated content progressively deteriorates when models are recursively trained on AI-generated data.[4] These academic endeavor has attracted attention from major media outlets such as Forbes[5], The New York Times[6], and the Harvard Business Review[7], all of which have reported extensively on the risks of relying too heavily on AI-generated data.

This paper examines the economic implications of the rise of generative AI and its created data,

[1] The report is available here.

[2] This shift is already underway: Amazon uses synthetic, AI-generated data to train its Alexa language assistant, while Apple has trained its Mm1 model using data generated from GPT-4 and LLaMA. The synthetic data industry is experiencing rapid growth, with market projections showing an expansion from 381.3 million in 2022 to 2.1 billion by 2028, representing an annual growth rate of 33.1%.

[3] The table only covers works between May 2023 and July 2024. However, the literature is experiencing rapid growth, and newer publications beyond this period are not included. Interested readers are encouraged to explore the subsequent developments independently.

[4] These academic findings are consistent with industry observations, such as the 2023 GitClear report that identifies a "downward pressure" on the quality of AI-generated code used with GitHub Copilot.The report is available here.

[5] https://www.forbes.com/sites/bernardmarr/2024/08/19/why-ai-models-are-collapsing-and-what-it-means-for-the-future-of-technology/

[6] https://www.nytimes.com/2024/07/19/technology/ai-data-restrictions.html

[7] https://hbr.org/2023/11/has-generative-ai-peaked

particularly focusing on AI-human competition and potential regulation. The starting point is to develop an economic framework to understand the interactions between AI adoption and data quality. This framework differs from existing literature, which typically treats AI as a new form of automation without considering the role of data (Aghion et al., 2019; Korinek and Suh, 2024; Acemoglu, 2024). Instead, we build a social-learning model with endogenous AI adoption where data is valuable because it helps predict some economic fundamentals which are not directly observable and are time-varying (Farboodi and Veldkamp, 2022). In the model, agents choose between using AI or human labor for production, and their production behavior generates data, potentially of different quality. Here, quality refers to how much *new* information the data adds to existing datasets for predicting future economic fundamental, as repetitive information is not useful. The key productive advantage of AI over human labor is its ability to access and handle "big data"—the rich historical dataset that records all past economic activities, whether performed by humans or AI. Since AI and human-generated data may differ in quality, the *composition* of economic activities between AI and humans affects overall data quality and, consequently, AI productivity. This creates an AI-data feedback loop: AI productivity influences the choice between AI and human labor, which in turn affects both the share of AI production and the mix of human and AI-created data, which determines AI productivity.

The model delivers the following theoretical insight: because of its access to the rich historical dataset, AI can have *higher* productivity compared to human labor while creating data of *lower* quality. In a sense, having access to the big data acts as a double-edged sword because, while boosting AI productivity, it also means AI's actions largely mimic past human behaviors instead of contributing truly novel information. In contrast, humans, lacking easy access to historical datasets, must often take more innovative action that, while potentially less productive, could add more valuable new information to the overall dataset. As an example, consider planning a trip to somewhere you have ever been before, for example, Ecuador. ChatGPT can quickly create a high-quality itinerary based on its database of past travels done by other people. However, it can only recommend places already in its database—places others have visited before. If you plan the trip yourself, you might miss some famous sites due to lack of information, but you could, through self-exploration, discover hidden gems like a local breakfast spot serving amazing bagels to local residents only. Once shared online, your discovery adds valuable new information to the overall database, helping future AI to make travel recommendations. If everyone solely followed ChatGPT's suggestions, such hidden gems might never be found.

Because of the AI-data feedback loop, the model features rich dynamics, such as multiplicity, stability, and cyclicality. The model dynamics depend crucially on one aspect of the model: the relative quality of AI versus human-generated data. Multiple steady states can arise when AI-generated data far exceeds human-generated data in quality. In one steady state, low data quality leads to low AI productivity, resulting in low AI adoption. Because human-created data is of lower quality, this confirms the low data quality in this steady state. In another steady state, high data quality drives high AI productivity and widespread AI adoption. The abundance of high-quality AI-generated data then sustains the steady state with high overall data quality. When AI-generated data is significantly lower in quality than human-generated data, the model exhibits cyclicality: high data quality today leads to high AI adoption, but the low quality of AI-generated data reduces overall data quality tomorrow, leading to lower AI adoption, and the cycle continues. Both the multiplicity and cyclicality implies some form of instability when data quality is highly elastic to AI adoption.[8]

To evaluate how the rise of AI can affect the economy, we model a one-time permanent reduction in AI costs in two scenarios. Our first experiment, the "AlphaGo experiment," explores what happens when AI-generated data is far superior in quality to human-generated data, allowing AI models to improve through "self-play" (learning from their own generated data). In this case, even a small reduction in AI costs can trigger a dramatic shift from zero to complete AI adoption. This is because of the self-reinforcement loop that more AI adoption leads to higher data quality, which leads to even more AI adoption. This transition features a rapid increase in AI-generated data and massive improvements in AI productivity, consistent with the real-world evolution from the original AlphaGo, which relied on human data, to AlphaGo Zero, which learned entirely through self-play (Silver et al., 2017; Holcomb et al., 2018). The key point of this experiment is that high-quality data generated by AI can lead to explosive growth of AI and economy, even if the model abstracts away from forces such as exogenous productivity growth or capital accumulation.

Our second experiment, the "ChatGPT experiment," examines the current rise of generative AI. To quantify the relative quality of AI versus human-generated data, we draw on recent computer science research (summarized in Table 1). These studies evaluate how foundational AI models—including large language models and image generation tools—perform when trained on AI-generated data. They consistently find quality deterioration when AI models are recursively

[8]When AI and human-generated data are of similar quality, the model generally converges to a unique steady state and achieves stability.

trained on AI-generated data. Based on this research, our calibration suggests that AI generates new data (or ideas) at only 8% of the quality of human-generated data. This substantial gap is identified with a sharp decline in AI quality when synthetic, AI-generated data enters the training process of AI (Chen et al. 2023; Alemohammad et al. 2023; Martínez et al. 2023). Our estimation indicates that the rise of AI-generated data imposes significant risks of corrupting future datasets in those tasks undertaken by generative AI.

Given this risk of data corruption, our model suggests that concerns about labor displacement of AI may not be as bad as you think. To illustrate this, we simulate model dynamics following a one-time stylised cost reduction shock which causes AI adoption rate to spike to 100%, completely displacing human labor in the short term. However, as more AI-generated data accumulates, its lower quality gradually degrades the training dataset and thus AI productivity. Over the long run, labor displacement is partially reversed due to the declined AI productivity and the economy stabilizes with AI handling about 70% of total production. This suggests that data quality concerns can reverse roughly 30% of the short-run labor displacement over time.

The fact that labor displacement is less concerning does not alleviate the need for government intervention. The key reason for intervention is a data externality: individuals don't consider how their private actions affect the quality of data available to everyone else. Our model examines two crucial aspects of private decision-making that might need regulation: AI adoption rates and AI's access to historical data. Both issues are central to current policy debates. The European Union Artificial Intelligence Act, the first comprehensive AI law, addresses both aspects through regulations on AI usage and data governance. Canada's Artificial Intelligence and Data Act (AIDA) proposes similar requirements. Our model shows that when AI-generated data reduces overall data quality, taxing AI adoption can help correct this negative externality.[9]

We also explore whether limiting AI's access to datasets improves welfare. A common justification for such regulations is to protect privacy and ensure safety and fairness in AI systems. Beyond these ethical concerns, this paper highlights an additional efficiency benefit of constraining AI's access to data: restricting data access "forces" AI to generate new information rather than relying on existing data, potentially improving the quality of AI-generated content. Our model suggests that as data corruption becomes more severe, the costs of restricting AI's data access decrease over time, strengthening the case for such restrictions.[10]

[9] If greater AI adoption improves overall data quality, our model suggest a subsidy to AI adoption.

[10] In principle, allowing agents to trade data could address this externality problem through market mechanisms (see Farboodi and Veldkamp (2022)). However, since data quality is both endogenous in the model and perhaps hard to

The paper connects to the literature that investigates the influence of data on the macroeconomy, as discussed in the works of Farboodi et al. (2019); Jones and Tonetti (2020); Cong et al. (2021); Abis and Veldkamp (2021); Cong et al. (2022); Farboodi and Veldkamp (2022). The paper proposes a novel modeling approach that jointly determines data quality and AI adoption. It further provides insights into policy implications and the regulation of AI to manage the quality of data efficiently.

The paper also relates to recent research on the impact of automation on labor, as explored by Acemoglu and Restrepo (2018, 2019, 2022); Acemoglu et al. (2022); Moll et al. (2022); Alonso et al. (2022); Jones (2023); Korinek and Suh (2024) and Acemoglu (2024). In this context, the paper considers the endogeneity of AI productivity with respect to data quality. The novel AI-data feedback loop allows the model to deliver new implications for labor displacement effects across various tasks with different levels of data quality. It is also the first paper that introduces direct experimental evidence from the computer science literature to discipline an economic model of AI.

The paper is linked to the literature on information economics and social learning, as explored by Morris and Shin (2002); Veldkamp (2005); Amador and Weill (2010); Ordonez (2013) and Fajgelbaum et al. (2017). Its key contribution is the application of a social learning framework to analyze the issue of AI adoption. It builds on the insights in this literature that more economic activities lead to more data being generated and hencebetter data quality. This paper further argue that the *composition* of data from different sources, i.e. the different types of activities conducted by AI v.s. human, matters for aggregate data quality.

2 Model

2.1 Primitives, Technology, and Friction

Time is assumed to be infinite and runs from $-\infty$ to $+\infty$. The economy is populated by $\bar{N}$ number of entrepreneurs and a representative household. Each entrepreneur has a lifespan of 1 period and is replaced by a newborn entrepreneur at the start of the next period.[11] These entrepreneurs have

observe in reality, a data market would likely face severe adverse selection problems, potentially leading to market failure as in Akerlof (1970). We therefore assume away such a market in the model. While beyond our current scope, exploring a frictional data markets could be a promising direction for future research.

[11] The assumption of short-lived entrepreneurs is made to maintain analytical tractability of the model. Extending the current framework to one with long-lived entrepreneurs would involve addressing information heterogeneity due to data hoarding.

the ability to produce a good of quality A_t^i, which they can sell to the representative household at a price of pA_t^i. The representative household is assumed with a utility function that is linear in the quality A_t^i, and for convenience, we normalize the slope of this linear relationship to 1. Therefore, we have $p = 1$.

The quality of production A_t^i is endogenous, and is influenced by entrepreneurs' action a_t^i. Specifically, I adopt the following functional form as in Morris and Shin (2002) and Farboodi and Veldkamp (2022):[12]

$$A_t^i = \bar{A} - \left(\theta_t - a_t^i\right)^2 \tag{1}$$

Where $\bar{A}$ is the maximum quality that can be achieved by some optimal action θ_t.

The variable θ_t represents the underlying economic fundamental of this economy. It is assumed that θ_t evolves according to

$$\theta_t = \rho\theta_{t-1} + \eta_t \tag{2}$$

where η_t is a noise with mean 0 and precision γ_η.

There is an information friction in the model: θ_t is not directly observable to any participants. This information frictions is the only friction in the model. Absent this friction, enterprenuers would always take actions alligned with the fundamental and the equilibrium achieves the first best allocation. θ_t can be interpreted as reflecting general economic conditions, market demand, or some optimal approach to research and development that is not easily available to individuals and requires efforts to obtain valuable information.

2.2 Timing, Information, and the Firm's Problem

Each entrepreneur upon his birth can choose to produce with its own labor or with an AI technology. Labor production would involve an idiosyncratic labor cost of φ_t^i while production with AI incures a capital cost of R_t. Besides the cost differences, labor and AI production also differ in the way they predict the economic fundamental and, hence, take the actions.

Specifically, at the start of period t, entrepreneurs are born with their individual labor cost φ_t^i, which is drawn from a distribution $F(\varphi)$. Then the fundamental θ_t is realized, together with a

[12] Farboodi and Veldkamp (2022) also explore a general setting under which the quality is given by:

$$A_t^i = g\left[\left(\theta_t - a_t^i\right)^2\right],$$

where g is a differentiable and strictly decreasing function. All of this paper's result goes through with this generalization.

public signal that is available to all individuals:

$$S_t = \theta_t + \varepsilon_t^S$$

Here, ε_t^S represents noise in the public signal with a mean of 0 and precision γ_S. The parameter γ_S captures the level of aggregate uncertainty.

Labor Production: If the entrepreneurs choose to use their own labor, a private signal is generated:

$$s_t^{Li} = \theta_t + \varepsilon_t^{Li}. \tag{3}$$

The precision of this private signal is denoted as γ_L. The signal can be thought of as generated by information acquisition of the entrepreneur or obtained through "learning-by-doing" when the entrepreneur is conducting the task with his own labor. Based on this signal, the entrepreneur can produce the good using human labor at the cost of φ_t^i. One can interpret the cost as encompassing both the expenses associated with information acquisition and the production cost of labor. The entrepreneurs' optimal actions are determined by:

$$W^L(S_t, s_t^{Li}) = \max_{a_t^i} E(A_t^i(a_t^i)|S_t, s_t^{Li}) \tag{4}$$

The payoff from labor production, given the cost, can be expressed as:

$$V^L(S_t, \varphi_t^i) = E(W^L(S_t, s_t^{Li})|S_t) - \varphi_t^i$$

Here, the superscript L indicates that the entrepreneurs are utilizing their own labor for production.

AI production: We now explore the case of entrepreneurs who adopt AI in their production processes. The adoption of AI incurs a cost, denoted as R_t. R_t is treated as an exogenous required rate of return, consistent with the perspective of Acemoglu and Restrepo (2018) and others, who view AI technology as a capital good. Once the entrepreneur purchases the AI product by paying R_t, the AI carries out an action a_t^i on behalf of the entrepreneur.

How does AI choose the action? The AI has access to "big data," which is an extensive dataset containing all past public signals and past actions taken by entrepreneurs. Mathematically, this can be represented as the following information set:

$$\Omega_{t-1} = \{S_{t-j}, D_{t-j}^i\}_{j=1,2,3,\ldots,-\infty}^{i=1,2,3,\ldots\bar{N}}$$

Here, S_{t-j} denotes past public signals in period $t-j$, and D_{t-j}^i represents data points regarding past actions taken by entrepreneur i in period $t-j$. The process of how this data is generated will be discussed in the following section.

In addition to observing historical data, the AI is also capable of generating its own private information, which can be expressed as:

$$s_t^{Ai} = \theta_t + \varepsilon_t^{Ai}, \varepsilon_t^{Ai} \sim N(0, 1/\gamma_A)$$

The use of large datasets incorporating past data (Ω_{t-1}) to predict future outcomes is a common practice in many applications of machine learning. For example, retail firms may utilize historical data to estimate future market demand, and hospitals might employ data from past patient records to make diagnoses. In addition to historical data, AI can also create its own, novel information and act upon it. This captures AI's "creativity" or its ability to "innovate", in addition to replicating what people did in the past. The quality of such new information is given by γ_A. If $\gamma_A \to 0$, AI in the model converges to a pure machine learning tool while if $\gamma_A \to \infty$, AI's action would only depend on its newly generated information and not on historical past data.

It is important to note that the emphasis on prediction versus new information generation may vary across different tasks. In certain areas chess and Go-playing, where the reward objective and the environment is relatively straightforward and deterministic, or in research settings where AI can explore novel drug combinations in high-dimensional spaces, the generation of new information by AI could be highly valuable. For instance, the AlphaGo software is able to significantly improve upon itself through self-play (Silver et al., 2017). Conversely, in a market context where AI is used to estimate demand of a product in some local markets, prediction based on past data might play a more significant role. The model accommodates both dimensions. In the numerical section we attempt to use experimental evidence from Large language models and image generation AI tools to assess the relative importance of both dimensions in those tasks.

Based on both sources of data, the AI picks an action to maximizes the value of the entrepreneur:

$$W^A(S_t, s_t^{Ai}, \Omega_{t-1}) = \max_{a_t^i} E(A_t^i(a_t^i)|S_t, s_t^{Ai}, \Omega_{t-1}) \tag{5}$$

The payoff from AI adoption is then given by:

$$V^A(S_t) = E(W^A(S_t, s_t^{Ai}, \Omega_{t-1})|S_t) - R_t$$

Here, the superscript A indicates that the entrepreneur has adopted AI for production.

Given the value functions, the agent selects the option that results in the highest ex-ante value:

$$\max\{V^l(S_t, \varphi_t^i), V^A(S_t)\}$$

Due to the monotonicity of $V^l(S_t, \varphi_t^i)$ with respect to φ_t^i, there exists a cutoff value denoted as $\bar{\varphi}$ such that agents adopt AI only if their individual φ_t^i exceeds this cutoff. This determines the num-

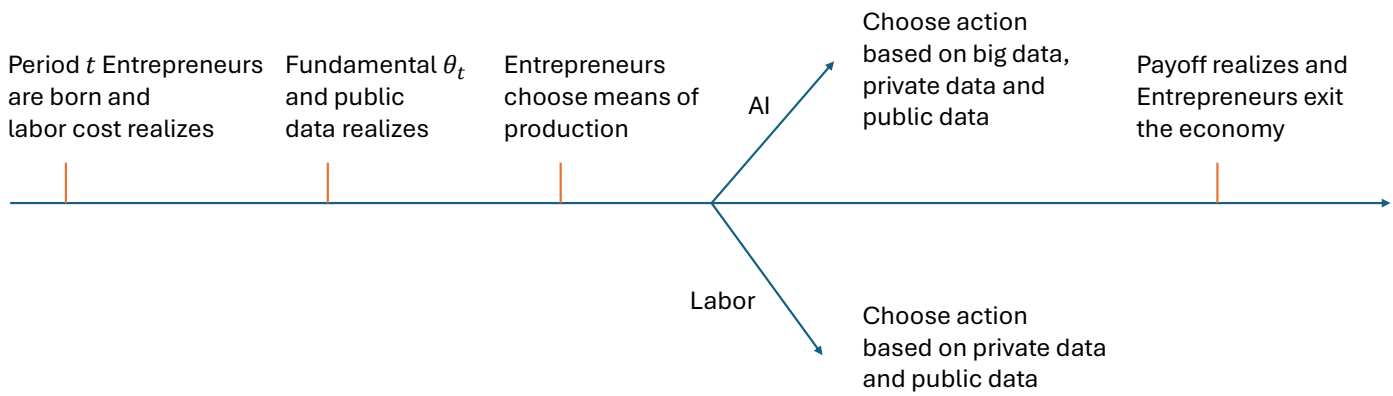

Figure 1: Timeline within a Period

ber of labor-adopting entrepreneurs, denoted by N_t. The number of AI-adopting entrepreneurs is thus given by $\bar{N} - N_t$. After production, the entrepreneurs exit the economy and are replaced by a new group of entrepreneurs in the following period. The timeline is summarized as in figure 1. Note that the only inter-period linkage is informational: that past actions become the data that feed into future versions of AI.

Remark on AI-Labor complementarity/substitutability: The relationship between labor and AI (or more generally, automation) has been an important topic in recent economic research, with studies suggesting that AI can act as either a complement or a substitute for labor, depending on the nature of tasks. AI often complements non-routine cognitive tasks and creative work, while substituting for routine and repetitive tasks that are easier to automate. To focus on the AI-data tradeoff, this paper adopts an agnostic approach on this issue, assuming that an individual's payoff function does not depend on the AI adoption decisions of others. As a result, the model abstracts from action complementarity or substitutability, to focus on the interactions from information linkages across agents. Extending the model to incorporate rich strategic responses to AI adoption would be a fruitful avenue for future research.

2.3 Data Generation Process

We now discuss the data generating process and how data points D_t^i are related to economic fundamental. When an entrepreneur takes an action, a data point is recorded as a noisy signal about that action:

$$D_t^i = a_t^i + \varepsilon_t^{Di}. \tag{6}$$

Here, the noise $\varepsilon_t^{Di} \sim N(0, \bar{N}/\gamma_D)$ can be interpreted as the information loss in the data collection process.[13]

In our model, we make the assumption that actions are directly observable, while private information is not. This assumption is commonly found in the social learning literature (see, for example, Fajgelbaum et al. 2017) and is consistent with real-world observations where companies consider their data as valuable assets and refrain from sharing it with other entities. For instance, Facebook's private data is not directly visible to external parties, but its various business activities are, allowing outsiders to make partial inferences about the private data held by Facebook.[14]

Given this setup, the idiosyncratic data points in a given period can be summarized into the aggregate means that reflect information generated by human labor and AI, respectively:

$$X_t^l = \frac{1}{N_t} \sum_{i \leq N_t} D_t^i \tag{7}$$

$$X_t^A = \frac{1}{\bar{N} - N_t} \sum_{i > N_t} D_t^i \tag{8}$$

Here, N_t represents the number of entrepreneurs adopting labor, and hence $\bar{N} - N_t$ is the number of entrepreneurs adopting AI. Thus, the overall information set available for AI at time t is:

$$\Omega_{t-1} = \{S_{t-j}, X_{t-j}^l, X_{t-j}^A\}_{j=1,2,3,\ldots,-\infty} \tag{9}$$

[13]Following Fajgelbaum et al. (2017), we assume that the precision of these data points D_t^i are inversely proportional to $\bar{N}$. This prevents the signals from being fully revealing as $\bar{N}$ tends to infinity. This assumption captures the notion that noise persists even with big data due to the increasing complexity of large economies and the challenges of handling larger datasets with constrained computational power.

[14]In the model we do not allow agents to trade their private information or sell their data in a marketplace. Since data quality is heterogeneous and perhaps hard to observe in reality, a data market would likely face severe adverse selection problems, potentially leading to market failure as in Akerlof (1970). We therefore assume away such a market in the model. Exploring a frictional data markets could be a promising direction for future research, see Farboodi and Veldkamp (2022) for an example of frictionless data market.

2.4 Equilibrium

We now define an equilibrium. The information equilibrium is competitive in the sense that each agent is atomic and takes the evolution of aggregate data as given. The data then evolves consistent with each agent's choices and actions.

Definition 2.1 *An information equilibrium, given sequences of exogenous shocks and parameters, consists of a sequence of individual decision rules* $\left\{\mathbb{1}^i_{At}, a^i_t\right\}^{i=1,2,\ldots\bar{N}}_{t=-\infty,\ldots,+\infty}$*, a sequence of aggregate labor-adopting entrepreneurs* $\{N_t\}_{t=-\infty,\ldots,+\infty}$*, a sequence of data points* $\left\{X^l_t, X^A_t\right\}_{t=-\infty,\ldots,+\infty}$ *and a sequence of beliefs* $\{\Omega_t\}_{t=-\infty,\ldots,+\infty}$ *such that*

1. $\left\{\mathbb{1}^i_{At}, a^i_t\right\}^{i=1,2,\ldots\bar{N}}_{t=-\infty,\ldots,+\infty}$ *solves individual entrepreneur's problem given the sequence of beliefs.*

2. $\{N_t\}_{t=-\infty,\ldots,+\infty}$ *is given by* $N_t = \sum_{i=1}^{\bar{N}} \mathbb{1}^i_{At}, \forall t$.

3. $\left\{X^l_t, X^A_t\right\}_{t=-\infty,\ldots,+\infty}$ *are given by:*

$$X^l_t = \frac{1}{N_t} \sum_{i \leq N_t} \left(a^i_t + \varepsilon^{Di}_t\right)$$

$$X^A_t = \frac{1}{\bar{N} - N_t} \sum_{i > N_t} \left(a^i_t + \varepsilon^{Di}_t\right)$$

4. *Information evolve according to:*

$$\Omega_t = \{S_{t-j}, X^l_{t-j}, X^A_{t-j}\}_{j=0,1,2,3,\ldots,-\infty}, \forall t$$

3 Characterization

Our focus will be on the limiting case as the number of entrepreneurs in the economy, $\bar{N}$, tends to infinity. In this scenario, the number of labor-adopting agents versus AI-adopting agents becomes deterministic. The share of labor-adopting entrepreneurs converges to the function $F(\bar{\varphi}_t)$.

$$n_t = \frac{N_t}{\bar{N}} \to F(\bar{\varphi}_t)$$

Here, $\bar{\varphi}_t$ represents the threshold labor cost above which an entrepreneur would choose to adopt AI.

3.1 A Recursive Structure in Data Quality

To characterize the information equilibrium, we begin by identifying a recursive structure in the evolution of aggregate data accumulation. Following Farboodi and Veldkamp (2022), we measure

the quality of data by the conditional precision of fundamental given the dataset Ω_{t-1}:

$$\gamma_t = \frac{1}{Var\left(\theta_t | \Omega_{t-1}\right)} \tag{10}$$

It is worth noting that due to the linear quadratic form of the entrepreneurs' objective functions, the only state variable that needs to be tracked is the precision γ_t, rather than the mean.

We now derive a law of motion for γ_t. To do so, note that the information sets defined in equation 9 admit the following recursion:

$$\Omega_t = \Omega_{t-1} \cup \{S_t, X_t^l, X_t^A\}. \tag{11}$$

Thus to derive the law of motion for conditional beliefs we just need to see how beliefs evolve through S_t, X_t^l, and X_t^A. We turn now to this task.

3.2 Optimal Actions Human v.s. AI

We now derive the optimal action rules for human labor and for AI. Taking the first order condition with respect to equation 4 and 5, and exploit the quadratic structure of the objective function (equation 1), we obtain the following lemma for the optimal decision rules:

Lemma 3.1 *The optimal action for human labor depends on the public data and the private data:*

$$a_t^i = E(\theta_t | S_t, s_t^{Li}) = \frac{\gamma_S}{\gamma_S + \gamma_L} S_t + \underbrace{\frac{\gamma_L}{\gamma_S + \gamma_L} s_t^{Li}}_{\text{Information content in labor action}} . \tag{12}$$

The optimal decision for AI depends on the public data, historical data, and private data generated by AI:

$$a_t^i = E(\theta_t | \Omega_{t-1}, S_t, s_t^{Ai}) = \frac{\gamma_t}{\gamma_S + \gamma_A + \gamma_t} \mu_t + \frac{\gamma_S}{\gamma_S + \gamma_A + \gamma_t} S_t + \underbrace{\frac{\gamma_A}{\gamma_S + \gamma_A + \gamma_t} s_t^{Ai}}_{\text{Information content in AI action}} . \tag{13}$$

where γ_t is given by equation 10 and $\mu_t = E\left(\theta_t | \Omega_{t-1}\right)$ is the conditional mean of θ_t given historical dataset.

The optimal actions are a weighted average of those information perceived by human and AI, where the weights are functions of the different qualities (precision) of those information. For labor, this means taking a weighted average of public data S_t and labor-generated data s_t^{Li}. As the quality of labor-generated data increases, the action becomes more sensitive to it. For AI, its optimal action is a weighted average of public information, conditional mean from the historical data, and some private information. The higher the quality of historical data (denoted by γ_t), the more weight AI assigns to it, and consequently, less weight is placed on other sources of data, including AI's own privately generated data S_t^{Ai}. Note that, because both the public signal and

historical data are already part of the big dataset, the only valuable source of variation for both human an AI comes from its private information (highlighted in red).

Because of its access to historical data (γ_t), AI's action generally depends less on its privately generated information than labor's action does:

$$\frac{\gamma_A}{\gamma_S + \gamma_A + \gamma_t} < \frac{\gamma_L}{\gamma_S + \gamma_L} \tag{14}$$

as long as γ_A is not significantly greater than γ_L. While having access to historical data strengthens AI's ability to make "better" decisions, it also means that AI's actions are more reflective of past events (already recorded in the datasets) and less influenced by the newly-generated information AI could contribute to aggregate data quality.[15]

Corollary 3.1 *The AI's actions are less sensitive to its privately generated information compared to humans, given that the quality of its private information is not significantly higher than that generated by humans.*

3.3 Data Generation

We now turn to data generation, holding fixed the relative share of AI and labor adoption.

The labor signal X_t^L: given the optimal action rule derived in lemma 3.1, the human action generates a data point:

$$D_t^i = a_t^i + \varepsilon_t^{Di} = \underbrace{\frac{\gamma_S}{\gamma_S + \gamma_L} S_t}_{\text{Public Info.}} + \underbrace{\frac{\gamma_L}{\gamma_S + \gamma_L} s_t^{Li} + \varepsilon_t^{Di}}_{\text{Idiosyncratic Info.}}$$

Since S_t is publically known, observing D_t^i is equivalent to observing the idiosyncratic part $\frac{\gamma_L}{\gamma_S+\gamma_L} s_t^{Li} + \varepsilon_t^{Di}$, which is equivalent to the following signal of the fundamental θ_t:[16]

$$\theta_t + \varepsilon_t^{li} + \frac{\gamma_S + \gamma_L}{\gamma_L} \varepsilon_t^{Di}.$$

Summing up all the data points for human-generated data (equation 7):

$$X_t^l = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\theta_t + \varepsilon_t^{Li} + \frac{\gamma_S + \gamma_L}{\gamma_L} \varepsilon_t^{Di} \right)$$

which follows a normal distribution and has a mean equal to the fundamental θ_t, as all other noise terms have means equal to zero.

[15] Attentive readers may have noticed that equation 14 does not hold when the quality of the AI's private signal is extremely high, i.e., as $\gamma_A \to \infty$. In this scenario, the AI would place nearly 100% of the weight on its private data, effectively disregarding public datasets for training. This extreme case can be compared to the field of gaming, particularly when AlphaGo was introduced, where the AI was trained exclusively by playing against itself. The model suggests that, in such a case, the data generated by the AI's action would surpass the quality of human-generated data.

[16] To obtain this, multiply the idiosyncratic part $\frac{\gamma_L}{\gamma_S+\gamma_L} s_t^{li} + \varepsilon_t^{Di}$ by $\frac{\gamma_S+\gamma_L}{\gamma_L}$, and we obtain $s_t^{li} + \frac{\gamma_S+\gamma_L}{\gamma_L} \varepsilon_t^{Di}$. The signal expression can be obtain by noticing that $s_t^{li} = \theta_t + \varepsilon_t^{li}$ (equation 3).

Taking the number of entrepreneurs $\bar{N}$ to ∞, we obtain the following result:

Lemma 3.2 *The information contents in human production activities can be summarized in X_t^l which is an unbiased signal of the fundamental θ_t with precision $n_t \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D$.*

The quality (or precision) of the signal is affected by the share n_t, as a larger share of labor increases the amount of information generated by labor. The term $\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2$ reflects how agents' actions are weighted between the public data and the privately generated data. Since the public signal is observed by everyone, only the portion of the variation in actions that stems from the private signal (captured by $\frac{\gamma_L}{\gamma_S+\gamma_L}$, see equation 12) contributes additional information to the overall dataset.

<u>The AI signal X_t^A</u>: given the optimal action rule derived in lemma 3.1, the AI action generates the following data point:

$$D_t^i = a_t^i + \varepsilon_t^{Di} = \underbrace{\frac{\gamma_S}{\gamma_S+\gamma_A+\gamma_t} S_t}_{\text{Public Info.}} + \underbrace{\frac{\gamma_t}{\gamma_S+\gamma_A+\gamma_t} \mu_t}_{\text{Dataset Info.}} + \underbrace{\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t} s_t^{Ai} + \varepsilon_t^{Di}}_{\text{Idiosyncratic Info.}},$$

which is a weighted average of the public signal S_t, the information in the big dataset (summarized by the conditional mean μ_t), and the private information generated by AI s_t^{Ai} (and observed by others with a noise ε_t^{Di}). Given that the public signal and datasets are observable by all, only the part of the variations in actions that comes from the AI-generated private signal contributes and enhances the overall quality of the dataset. Hence we have:

Lemma 3.3 *The information contents in AI production activities can be summarized in X_t^A which is an biased signal of θ_t with precision $(1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2 \gamma_D$.*

The precision of the signal is influenced by the AI adoption share $1-n_t$, as greater adoption leads to more AI-generated data. The middle term $\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2$ reflects how agents' actions put certain weight on the public data and historical data, and those variations in actions do not add new information beyond what is already incorporated in the aggregate data. Hence, the only valuable source of information one can extract from AI' action comes from the variation due to its privately-generated data, to which the AI's action optimally assigns a weight of $\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}$.

Comparing Lemma 3.2 and Lemma 3.3, it is clear that the relative quality of human- and AI-generated data depend on two aspects. First, it depends on the relative shares of labor n_t and AI adoption $1-n_t$. Second, it depends on how much weight each type assigns to privately generated data. For labor, this weight is $\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2$, while for AI, it is $\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2$. Given a positive γ_t, it is clear that AI generally assigns a lower weight compared to labor if the quality of the privately

generated data is similar. This reflects the fact that AI can leverage large datasets from past actions, which dilutes the impact of the private signal on AI's actions.

Corollary 3.2 *AI-generated data contains less valuable information than human-generated data, provided the quality of AI's privately generated data is not significantly greater than that of human-generated data.*

3.4 The Evolution of Information and the Data Corruption Channel

We now derive the law of motion of the stock of knowledge γ_t. Utilizing the recursion (equation 11), we have:

$$Var\left(\theta_t|\Omega_t\right) \quad = \quad Var\left(\theta_t|\Omega_{t-1}, S_t, X_t^l, X_t^A\right)$$

From lemma 3.2 and 3.3, we obtain the law of motion for γ_t:

Lemma 3.4 *Given γ_t and n_t, γ_{t+1} is given by:*

$$\begin{aligned}\frac{1}{\gamma_{t+1}} &= Var\left(\theta_{t+1}|\Omega_t\right) \\ &= \rho^2 \frac{1}{\underbrace{\gamma_t}_{prec.\ of\ \Omega_{t-1}} + \underbrace{\gamma_s}_{prec.\ of\ S_t} + \underbrace{n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D}_{prec.\ of\ X_t^l} + \underbrace{(1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2\gamma_D}_{prec.\ of\ X_t^A}} + \frac{1}{\gamma_\eta} \end{aligned} \tag{15}$$

where the four components in the denominator captures the information content in Ω_{t-1}, S_t, X_t^l and X_t^A respectively. Note that this is a partial equilibrium result in the sense that the share of labor-adopting entrepreneurs n_t is holding fixed. It is nonetheless useful for us to understand how n_t could affect knowledge accumulation. We now analyze what happens to aggregate information quality γ_{t+1} when we change the relative composition of AI and human (i.e. varying the value of n_t):

Theorem 3.1 *Holding fixed current data quality γ_t, an increase in AI adoption (lower n_t) leads to lower data quality in the future (lower γ_{t+1}) if and only if the relatively quality of information generated by AI is not too much higher than those generated by labor:*

$$\frac{\gamma_A}{\gamma_L} < \frac{\gamma_S+\gamma_t}{\gamma_S}. \tag{16}$$

proof: It can be observed from equation 15 that perturbations of n_t affect γ_{t+1} through the relative magnitude of $\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D$ and $\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2\gamma_D$. If the former is greater than the latter, a decrease in n_t leads to a decrease in γ_{t+1}. Simplifying this condition yields equation 16.

This theorem presents the key result of the paper: with increased AI adoption, the overall quality of the data can decline, as a larger share of AI-generated data corrupts the dataset. This data corruption channel presents when the quality of AI-generated data, relative to human-generated data, is sufficiently low.

A perhaps unexpected implication of the theorem arises when the quality of private information produced by both AI and humans is identical, i.e., $\gamma_L = \gamma_A$. In this case, greater AI usage could harm future data quality, as condition 16 holds. But why would this occur if AI and humans generate private information of equal quality? In the context of social learning, agents infer information from others' actions, which are noisy reflections of their privately-generated information, as they also respond to other data sources. Human actions are influenced by the public signal S_t, while AI actions are predominantly influenced by historical data Ω_{t-1} in addition to the public signal S_t. Since AI processes more diverse data sources, it places less emphasis on its own privately-generated data than humans do. Consequently, AI activities contribute less valuable information to the aggregate data quality compared to human activities.

Another interesting implication of this theorem is that the lower the public uncertainty (higher γ_S), the lower the value on the right hand side of equation 16, hence the more likely that AI can increase the quality of aggregate data.

One can also show that higher quality historical data may impede further data quality improvement. From condition 16, it is clear that the higher the quality of historical data, (i.e., higher γ_t), the larger the value on the right-hand side of equation 16, suggesting that the adoption of AI is more likely to impede data quality improvement. This occurs because higher γ_t values lead AI to rely more on historical data, placing less weight on its own signal generation, thus contributing less new information to aggregate data quality. In cases where prior data quality is very high, AI adoption offers great returns. In such situation, we can expect a surge in AI adoption, which may suppress long-term data quality.

We now close the model by solving for the equilibrium value of n_t based on individual optimality conditions.

3.5 Closing the Model: Endogenizing n_t

From the quadratic objective function, the optimal action is simply the conditional expectation of the fundamental parameter θ_t. $a_t^i = E(\theta_t|\text{information set})$, where agents use all the available information to forecast the fundamental, and take actions accordingly. Specifically, we now derive

the ex-post payoff of the entrepreneur who adopts labor to produce (equation 4):

$$W^l\left(S_t, s_t^{li}\right) = \bar{A} - \max_{a_t^i} E\left(\left(\theta_t - a_t^i\right)^2 | S_t, s_t^{li}\right)$$

Take first order condition with respect to a_t^i, we obtain:

$$a_t^i = E\left(\theta_t | S_t, s_t^{l,i}\right)$$

Plug it back into $W^l\left(S_t, s_t^{li}\right)$:

$$\begin{aligned} W^l\left(S_t, s_t^{li}\right) &= \bar{A} - Var\left(\theta_t | S_t, s_t^{li}\right) \\ &= \bar{A} - \frac{1}{\gamma_S + \gamma_L} \end{aligned}$$

where the last equality follows because the precision of the public signal and private signal are given by γ_S and γ_L respectively, and that these two signals are uncorrelated. Given that $W^l\left(S_t, s_t^{li}\right)$ does not depend on the specific realization of S_t, but only the second moments, we can derive the ex-ante payoff for labor adoption (equation 2.2):

$$\begin{aligned} V^l\left(\varphi_t^i\right) &= E\left(W^l\left(S_t, s_t^{li}\right) | S_t\right) - \varphi_t^i \\ &= \bar{A} - \frac{1}{\gamma_S + \gamma_L} - \varphi_t^i \end{aligned} \tag{17}$$

Similarly we can derive the payoff of adopting AI technology

$$\begin{aligned} V^A\left(\gamma_t\right) &= \bar{A} - Var\left(\theta_t | S_t, \Omega_{t-1}, s_t^{Ai}\right) \\ &= \bar{A} - \frac{1}{\gamma_S + \gamma_A + \gamma_t} - R_t \end{aligned} \tag{18}$$

Hence the threshold that determines the share of entrepreneur using each technology is pinned down by equalizing the payoffs in those two cases:

$$V^l\left(\bar{\varphi}_t\right) = V^A\left(\gamma_t\right)$$

Plugging in the expressions for the two payoff functions:

$$\bar{A} - \frac{1}{\gamma_S + \gamma_L} - \bar{\varphi}_t = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \gamma_t} - R_t$$

This gives the following threshold as a function of current stock of knowledge:

$$\bar{\varphi}_t\left(\gamma_t\right) = R_t + \frac{1}{\gamma_S + \gamma_t + \gamma_A} - \frac{1}{\gamma_S + \gamma_L}$$

This equation, together with the equation characterizing the dynamic evolution of information, pins down the information equilibrium in this model:

Theorem 3.2 *The equilibrium of the model is unique. The dynamics of the model is fully characterized by*

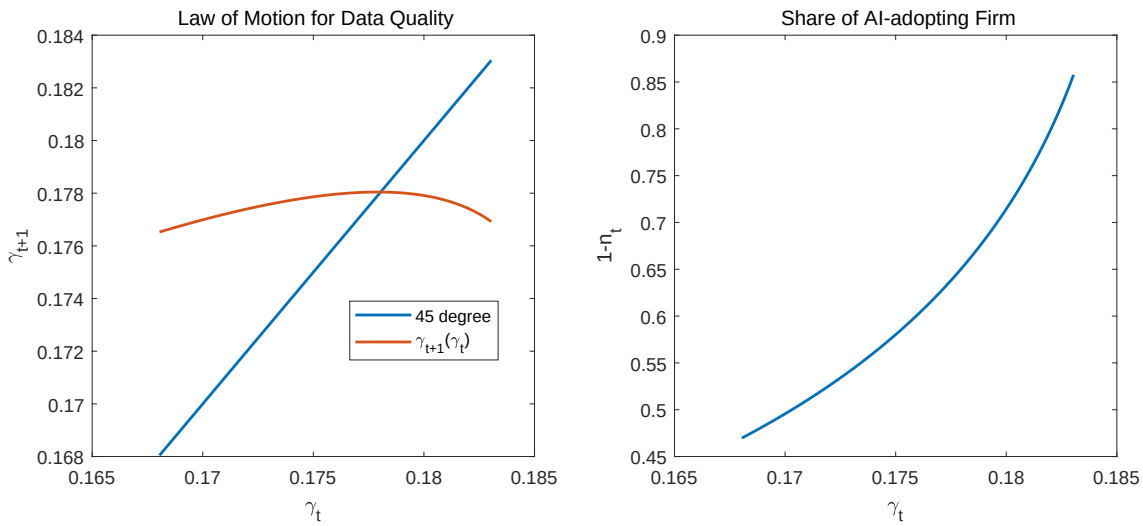

Figure 2: Law of Motion and Policy Functions

the following system:

$$\frac{1}{\gamma_{t+1}} = \rho^2 \frac{1}{\gamma_t + \gamma_s + n_t(\gamma_t)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1 - n_t(\gamma_t))\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta} \tag{19}$$

where the share of labor-adopting entrepreneurs is given by:

$$n_t(\gamma_t) = F(\bar{\varphi}_t(\gamma_t)) = F\left(R_t + \frac{1}{\gamma_S + \gamma_t + \gamma_A} - \frac{1}{\gamma_S + \gamma_L}\right),$$

where $F(.)$ is the cumulative distribution function for the labor cost and R_t is the cost of AI in period t.

Note that the data quality parameter γ_t appears in three different locations within the law of motion equation 19. Firstly, it embodies information passed down from previous periods. Secondly, it influences the equilibrium level of AI adoption represented by $1 - n_t$. Lastly, it also impacts the information generated through AI-associated activities because it alters the relative weight AI places on its privately-produced signal.

Figure 2 plots the equilibrium functions, including the law of motion for γ_t and the AI adoption share as a function of γ_t.[17] The left panel displays the law of motion for γ_t, which exhibits a non-linear trend, starting with an increase and then a decrease. The initial increase is due to more information from the past implying more information for the future, which is the standard information transmission effect. The decreasing part indicates that an abundance of information encourages greater AI adoption (as illustrated in the right panel). Increased AI adoption can, in

[17]Parameters used: $\rho = 0.95, \gamma_S = 0.05, \gamma_D = 0.1, \gamma_L = 0.1, \gamma_A = 0.1, \gamma_\eta = 0.5, R_t = 3.68, \forall t$. The function F follows a log normal distribution with a mean of -2 and a standard deviation of 2.

turn, reduce the quality of aggregate data.[18]

When the precision of the AI-generated signal γ_A approaches infinity, the model simplifies to one where the relative productivity of AI and labor remains constant, similar to Acemoglu and Restrepo (2018). Under this circumstance, AI can near perfectly predict θ_t using only the private signal, making AI's productivity consistently equal to the upper limit $\bar{A}$. For labor-adopting entrepreneurs, their expected productivity equals $\bar{A} - \frac{1}{\gamma_S+\gamma_l}$. Hence, the relative (expected) productivity of labor versus AI, using quality of the product produced as a measure of producitivty, is given by

$$\frac{\bar{A} - \frac{1}{\gamma_S+\gamma_L}}{\bar{A}}$$

which is a constant and independent of γ_t.[19]

More generally, this relative productivity is given by

$$\frac{\bar{A} - \frac{1}{\gamma_S+\gamma_L}}{\bar{A} - \frac{1}{\gamma_S+\gamma_A+\gamma_t}} \tag{20}$$

from which it is evident that a higher data quality (higher γ_t) results in AI being relatively more productive than labor, leading to more AI adoption. But note that this ratio is always less than 1, indicating that AI is always more productive than labor, given its access to richer sources of historical data.

Theorem 3.3 *When historical datasets are useful: $\gamma_t > 0$ and the relative quality of AI v.s. human signal satisfies:*

$$\frac{\gamma_L - \gamma_t}{\gamma_L} < \frac{\gamma_A}{\gamma_L} < \frac{\gamma_S + \gamma_t}{\gamma_S}, \tag{21}$$

AI is more productive than humans, but its action generates data of lower quality.

Note that the lower bound of $\frac{\gamma_A}{\gamma_L}$ in condition 21 is less than 1, while the upper bound is greater than 1, ensuring the interval is non-empty. Within this interval, the quality of the AI signal is sufficient for AI productivity to surpass human productivity with AI's extra access to the historical dataset,[20] even if the quality of the AI signal itself can be lower than that of the human signal (note that the lower bound $\frac{\gamma_L-\gamma_t}{\gamma_L}$ is less than 1 when $\gamma_t > 0$). When the dataset quality is sufficiently

[18] It is true in this numerical example where γ_L=γ_A. This satisfies the condition in theorem 3.1 and hence more AI adoption hurts data quality.

[19] In Acemoglu and Restrepo (2018, page 1495, assumption 1), relative productivity is assumed to be fixed within a task.

[20] One can verify that satisfying the lower bound implies the relative productivity of humans versus AI (see equation 20) is less than 1, meaning AI is more productive than humans.

high ($\gamma_t > \gamma_L$), the lower bound is negative, meaning that AI can be more productive than human without generating any useful information independently.

AI's reliance on historical datasets also implies that it places less weight on data it generates privately. The upper bound in condition 21 ensures that the precision of a single piece of human data, given by $\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D$, is greater than the precision of a single piece of AI data, given by $\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2 \gamma_D$. This guarantees that the quality of AI-generated data is lower.[21] Above this upper bound, AI is both more productive and can generate higher-quality data.

The higher the quality of the historical dataset (indicated by larger values of γ_t), the lower the lower bound and the higher the upper bound specified in condition 21. This implies that while AI can utilize high-quality historical data to enhance its productivity, the process of learning from this data and acting upon it does not contribute new information to the collective dataset, and could "crowd out" new information generated through AI. Hence higher-quality historical datasets increase the likelihood that AI becomes more productive, but they also reduce the quality of the data generated through its actions.

3.6 Model Dynamics

Due to the interaction between AI adoption and data quality, the model displays rich dynamics over time. Depending on the relative quality of AI v.s. human data, we divide the analysis into the following three cases:

Case 1: high quality AI-generated data

In this case multiple steady states with different levels of aggregate data quality can arise. When data quality is low, AI is not very productive relative to human, hence AI adoption is low and therefore almost all new data are generated by human and thus, by assumption, of lower quality. This confirms the low aggregate data quality in this steady state. When the initial data quality is high, on the other hand, AI productivity is high, leading to many agents to switch to AI. This creates high-quality AI-generated data, which confirms the initial high data quality in equilibrium. The law of motion for data quality is depicted in the top left panel of figure 3.

Theorem 3.4 *Given that $\gamma_S > 0$ and $\gamma_A \to \infty$, the dynamic system in theorem 3.2 admits at least two steady states for a sufficiently compressed labor cost distribution $F(.)$. In those two steady states, the steady*

[21] It is important to distinguish between AI signal and AI-generated data. In the paper, AI signal refers to the quality of information privately generated by the AI, while AI-generated data refers to the observable information inferred from AI actions.

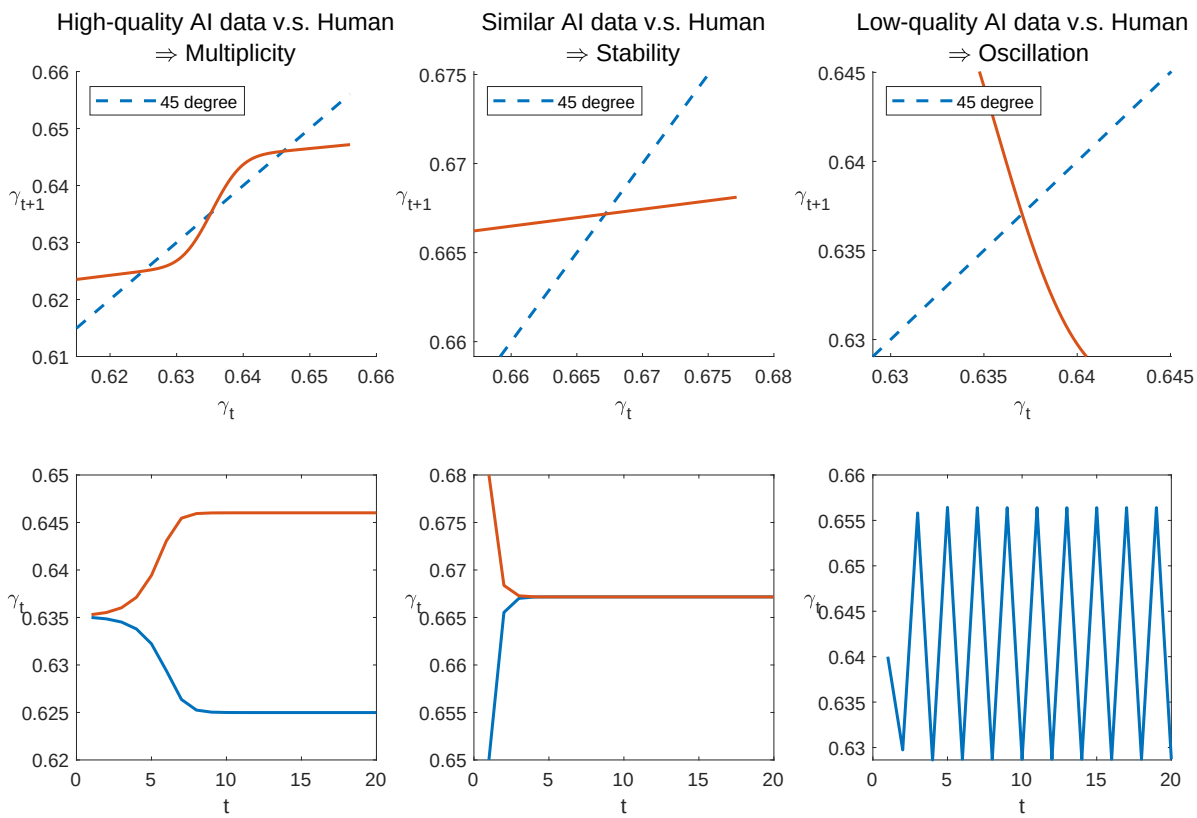

Figure 3: Model Dynamics –Multiplicity, Stability, and Cyclicality

state with high data quality is associated with high AI adoption rate.

In this case the equilibrium feedback loop between AI adoption and data quality is *strongly positive*: higher AI adoption leads to higher overall data quality, which leads to even greater share of AI adoption. This means that the derivative $\gamma_{t+1}(\gamma_t)$ has a steep and positive slope around the middle steady state (see top left panel) and hence multiple steady states can arise.

The multiplicity suggests certain fragility when AI data is of much higher quality compared to human-generated data. This is illustrated in the bottom left panel of figure 3: small variations in the initial condition of the model can create vastly different outcome over the long run, as they may converge to different steady states.

An AlphaGo Experiment: How Explosive Growth Happens

A natural application of this situation would be where AI-generated data is far superior in quality compared to human-generated data. The game of go would be a natural candidate as the the quality of AI can improve through self-play (using data generated by playing against itself). We start by parameterize the model. We perceive relatively little fundamental uncertainty, as the reward system and objective function of those games relatively deterministic compared to other real-world tasks which often involve greater uncertainty. We thus set $\rho = 0.999$, and $\gamma_\eta = 10$, indicating that the fundamental (optimal solution) hardly moves over time. Further increasing the value of ρ does not affect the simulation results. We then set the quality of the public information to be relatively low to capture that little is known (a priori) about the optimal solution: $\gamma_S = 0.01$. We also assume that AI-generated data can be highly useful in this setting, so that the quality of AI-generated data is 100 times more valuable than those generated by labor: $\gamma_A = 1$ and $\gamma_L = 0.01$. Finally, we set $\gamma_D = 10$ so that it is relatively easy to record data into the database for the training of AI.

The resulting law of motion for data quality is displayed as the red solid line in the top left panel of figure 4. One can see that the model displays multiple steady states where the low steady state no AI is adopted and a high steady state where there is 100 percent AI adoption.

We now simulate the model assuming that the world starts at the low steady state with zero AI adoption, which captures the long-held belief before Alphago emerged that computer can never defeat human player in the game of go. We consider a shock that reduces the cost of AI (R_t) by 5% in period 20. Such a reduction in cost shifts the law of motion for data quality up from the red solid line to the yellow dashed line (see top left panel of figure 4), and the low steady state disappears as a result. The equilibrium thus starts to transit to the high steady state. During this

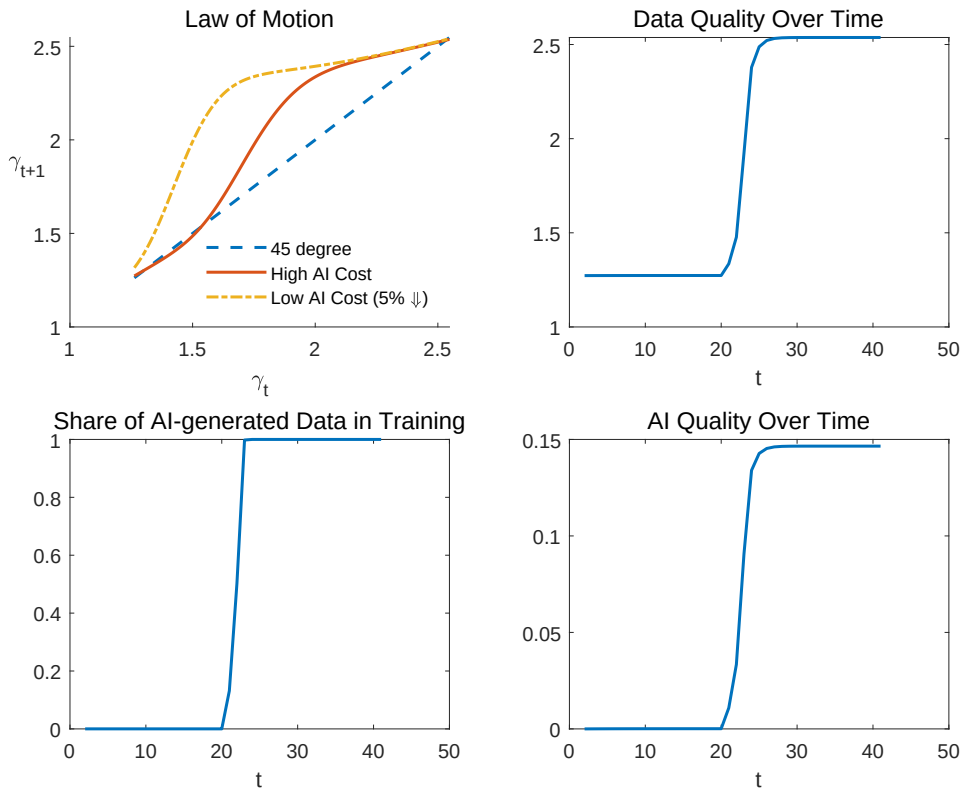

Figure 4: Model Dynamics – the AlphaGo Experiment

transition, the share of AI-generated data increases dramatically from 0 percent to 100 percent. This is consistent with the transition from AlphaGo (which still uses human inputs for training) to AlphaGo Zero, where all data input is generated by itself. As a result, aggregate data quality improves and AI productivity also increases dramatically within a relatively short time frame, which is consistent with the AlphaGo project formed in 2014 and have massively improved in relatively short time and when the AlphaGo was able to beat Lee Sedol in 2016 and world champion Ke Jie in 2017. Although framed in the context of Go-playing, the numerical example aims to capture something more general: rapid economic growth and productivity improvement where AI can "self-improve" through generating and consuming high-quality data. The explosive growth is similar to some types of the transformative technological progress such as those discussed in Trammell and Korinek (2023).

Case 2: Similar quality AI-generated data compared to human-generated data

In this case it is assumed that $\gamma_A = \gamma_L$. This implies that shifts in the share of AI adoption would not have a significant impact on overall data quality because the loadings on new informa-

tion are not so much different across humans and AI.[22] In such a case the equilibrium feedback loop between AI adoption and data quality is weak, and does not significantly affect the aggregate law of motion for γ_t. Hence the model displays stability with global convergence to the unique steady states, as a standard model of Bayesian learning. Results are displayed in the middle panel of figure 3.

Case 3: Low quality AI-generated data

In this case I set γ_A to be 10 percent of γ_L, suggesting that AI-generated data is of lower quality than labor. In this case, endogenous cycles can arise: higher data quality today leads to higher AI productivity and hence more AI adoption. This creates more low-quality AI-generated data and leads to lower overall data quality in the future. Thus the aggregate law of motion for γ_t can take a negative slope (top right panel of figure 3).

One can formally see the negative slope by examining the dynamic equation in theorem 3.1 in the special case where $\gamma_A \to 0$. In this case, AI is a pure machine learning tool and thus AI-generated data does not contribute to improvements in overall data quality. The law of motion for data becomes:

$$\frac{1}{\gamma_{t+1}} = \rho^2 \frac{1}{\gamma_t \uparrow + \gamma_s + n_t \downarrow (\gamma_t) \left(\frac{\gamma_L}{\gamma_S + \gamma_L}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta}$$

An increase in γ_t has a direct positive impact on γ_{t+1} but a negative impact through n_t because there are less fresh data generated through human production. The negative impact overpowers the positive direct effect if the derivative of $n'(.)$ is sufficiently large. This is guaranteed by a sufficiently compressed distribution of labor cost.

Theorem 3.5 *Given that $\gamma_A \to 0$, the law of motion for γ_t is downward slope at the steady state for a sufficiently compressed labor cost distribution.*

The implication of the downward-sloping law of motion is that it leads to endogenous cycles in both data quality and AI adoption. This is shown in the bottom right panel of figure 3, where the economy bounces around situations with high data quality and low data quality. The AI adoption rate would also exhibit similar oscillation as the AI Adoption rate is a monotonic function of overall data quality γ_t.

We conclude that the model can deliver rich dynamics depending on the quality of AI-generated data compared to human data. We now attempt to quantify this critical aspect of the model using

[22] The loadings of actions on private information won't be exactly the same between AI and human, but they are not so much different in this case.

experimental evidence from the computer science literature.

4 Quantification of Data Quality in Generative Tasks

We now quantify the model in the context of tasks conducted by current version of generative AI. Those tasks include producing a variety of novel content, such as images, video, music, speech, text, software code and product designs. The key aspect of the calibration is to discipline the relative quality of information generated by AI v.s. human: $\frac{\gamma_A}{\gamma_L}$ in those tasks, which will play a crucial role in determining model dynamics. We will now discuss the evidence to inform this aspect of the model.

4.1 FID Score and Fully Synthetic Loop

We consult the computer science literature for how to measure the quality of information produced by AI. A crucial metric in this assessment is the Fréchet Inception Distance (FID), a popular measure that gauges the similarity between outputs from a generative model and the data used to train it. Mathematically the FID score calculates the distance between two probability distributions. In practise, it effectively compares the empirical distribution of real data (e.g. images) with the synthetic distribution produced by AI. A lower FID score indicates a closer resemblance between these two distributions, and higher quality of AI. The FID score can be thought of as a loss function—where a higher score reflects lower AI quality.

We first calculate the FID score between the true value of fundamental θ_t and its conditional distribution given the information available to AI, Ω_{t-1}. Given that both distributions are normally distributed, we apply a specific formula for the FID score adapted to (single dimensional) normal distributions (Dowson and Landau, 1982), where the distance between two normal distributions with means μ_x, μ_y and standard deviations σ_x, σ_y is calculated as:

$$(\mu_x - \mu_y)^2 + (\sigma_x - \sigma_y)^2$$

We determine the average FID score in the model by averaging across a large panel of simulations. The reason for running the simulation is that the mean of the probability distributions could change due to exogenous shocks. By averaging across a large panel of simulations we can wash out the noises and obtain a robust measure of AI performance.

As discussed in the introduction, the emerging computer science literature on data corruption consistently finds that increasing reliance on synthetic data generated by AI can degrade data

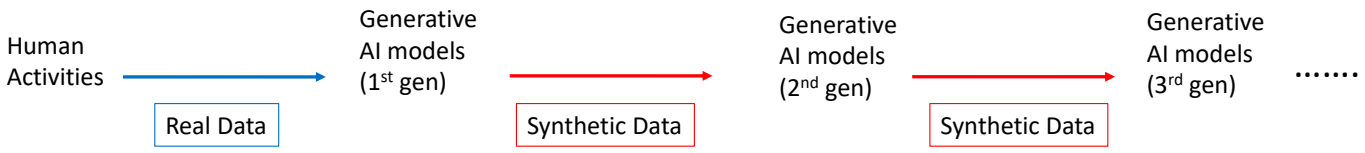

Figure 5: Fully Synthetic Loops

quality. In this literature the general methodology involves training generative AI software with real-world data and then using the synthetic data it produces to train subsequent AI generations, creating "fully synthetic loops." This process, illustrated in figure 5, repeats over several iterations, with each AI generation being trained exclusively on synthetic data produced by the preceding generation (Alemohammad et al., 2023).

The key finding from those studies is that the FID score tends to increase with each iteration of full synthetic loops, suggesting a decline in AI-generated data quality. For instance, Alemohammad et al. (2023, see their figure 8) observed over 400% increase in FID scores during full synthetic loops. Martinez et al. (2023, see their figure 3(c)) notes a rise of over 700%, and Bohacek and Farid (2023, figure 2) find an approximately 300% increase throughout full synthetic loops in a most recent study.

I argue that this pattern of increasing FID scores across synthetic loops can help identify a key parameter in our information model: the relative quality of information produced by AI compared to humans. Figure 6 demonstrates this by showing the relationship between the quality of AI-generated information and the FID score's trajectory, with a steeper increase in FID score implying faster detorioration of data quality. One can easily see that the lower the quality of information provided by AI, the more the FID score increases during full synthetic loops. Thus, faster growth of the FID score indicates lower quality of information provided by AI. In the calibration I thus match the most conservative estimate of the FID score increase provided by Bohacek and Farid (2023), which is 300%. This provides an upper bound for the relative information quality of AI and even at this upper bound the calibration indicates that the relative quality of information provided by AI is only 8% of that by human.

Our calibration results imply that AI contributes minimally to the generation of new information or knowledge beyond what is already encapsulated within the aggregate dataset. This finding aligns with the perspective that current version of generative AI fundamentally operates as a machine learning tool, which extrapolates predictions and insights based on pre-existing data and historical information. Consequently, while AI may surpass human capabilities in terms of productivity, this advantage does not translate into significant enhancements in the overall quality of the aggregated data. This suggests that the value added by AI in the context of generating novel insights or improving the informational quality of the dataset is limited. It underscores the notion that AI's primary function remains within optimizing and analyzing existing information rather than contributing original knowledge or fundamentally novel insights to the dataset.

4.2 Calibration of Other Parameters

I now describe how the remaining parameters are calibrated, beginning with the calibration of the fundamental process as defined in equation 2. The calibration is based on the autocorrelation and standard deviation of the logarithm of output, resulting in standard parameter values: $\rho = 0.99$ and $\gamma_\eta = 1/(0.055)^2$.[23] We set the standard deviation of investment cost to 0.0155, following Fajgelbaum et al. (2017). The mean of the cost distribution is not separately identified from the AI adoption cost R, and is therefore normalized to 0.

Next, we address the calibration of information parameters: $\gamma_A, \gamma_L, \gamma_D$, and γ_S. According to equation 16, it's important to note that the absolute values of γ_A and γ_L are not as critical as their ratio, which determines the quantitative strength of the data corruption channel. Hence, we normalize the precision of the labor signal to $\gamma_L = 100$, making the precision of the AI signal γ_A a direct indicator of relative information quality. As discussed in section 4.1 and based on the experimental findings by Bohacek and Farid (2023) showing that the FID score increases by 300% over a full synthetic loop, we set $\gamma_A = 8$.

For γ_D, which measures the information loss during the data collection process (indicating that data in our model merely serves as a noisy mirror of past economic activities as seen in equation 6), we calibrate it to match the initial data quality observed in the full synthetic loop 5. The initial quality of the dataset, trained by humans and assessed using the FID score, is established at 90 by Bohacek and Farid (2023). Aiming to match this values leads us to set γ_D=35.

[23] This implies that the standard deviation of the innovation η_t is 0.055, consistent with those used in existing literature.

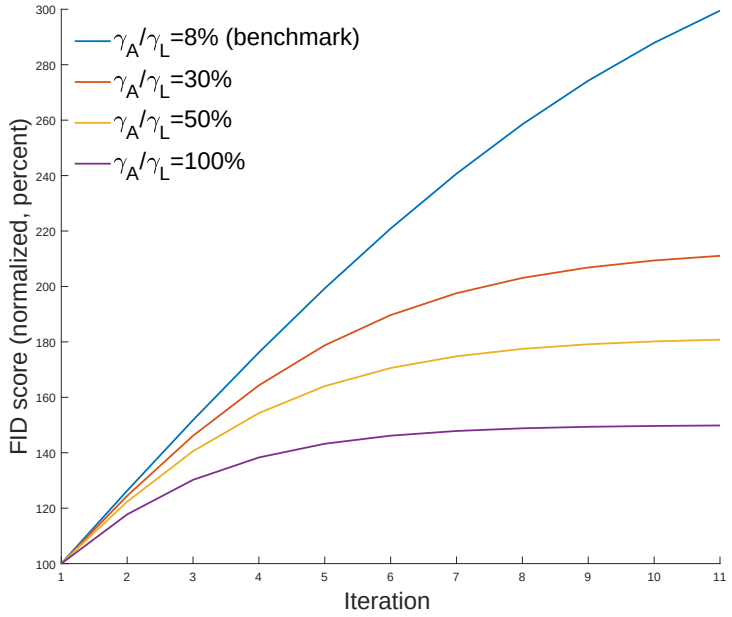

Figure 6: FID score in the model for full synthetic loops

Lastly, we assign $\gamma_S = 0$ due to the absence of additional targets for measuring the precision of public signals. However, in section 4.5, we conduct a sensitivity analysis to evaluate the impact of a higher γ_S value on our findings, to better understand how variations in this parameter could potentially influence the outcomes of our model. We find that our results doe not change with a changing precision of public signal, as long as the model is recalibrated to match the evolution of FID score during full synthetic loops as shown in section 4.1.

4.3 Quantitative Results: the ChatGPT Experiment

We will now describe the experiment designed to evaluate the long-run effects of AI adoption, termed the "ChatGPT" experiment. This experiment begins in a hypothetical world at time $t = 1$, where AI technology is either nonexistent or the costs associated with its adoption are prohibitively high (e.g. $R_0 \to \infty$). We then examine the model dynamics following a negative shock in period $t = 2$, which reduces the cost of AI to R_1. The cost stays constant after this period. This scenario could be interpreted as a significant technological breakthrough, such as the introduction of ChatGPT, which substantially lowers the barriers to AI accessibility and affordability.

Figure 7 displays the model dynamics. The top left panel plots the one-time permanent reduction in the cost of AI. Following this exogenous cost reduction, we analyze the model dynamics

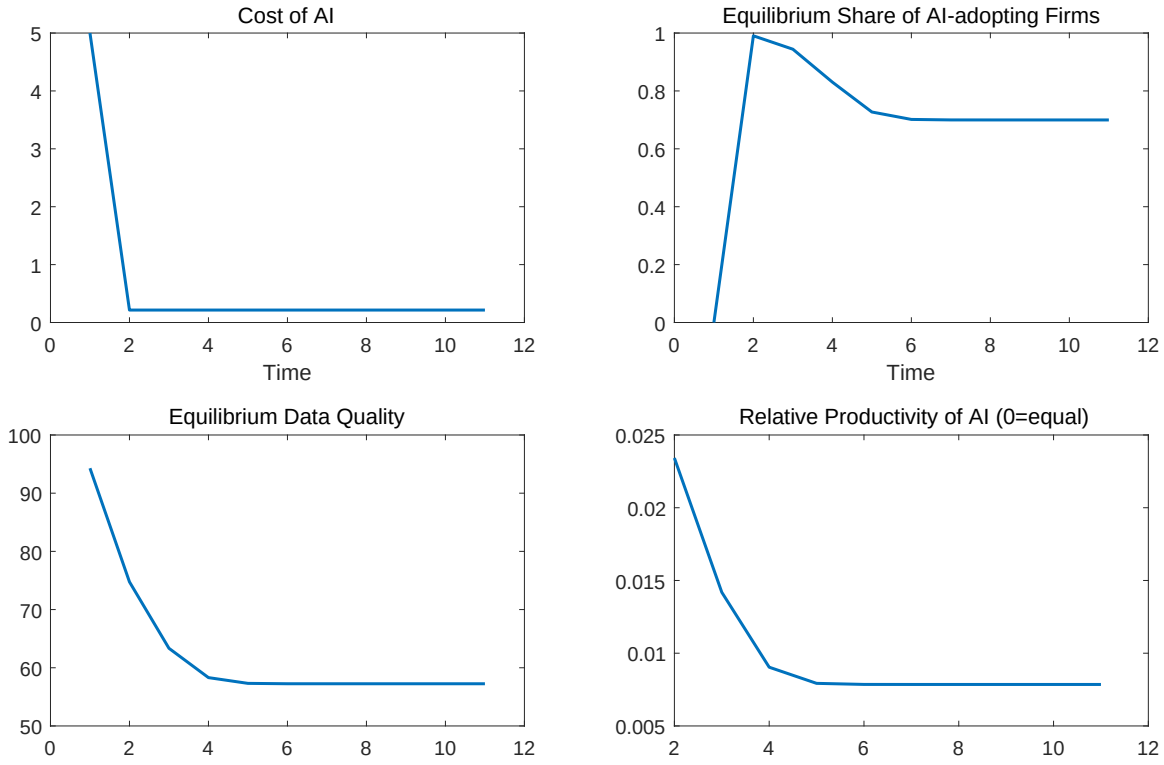

Figure 7: Model Dynamics – the ChatGPT Experiment

according to equation 19. The top right panel shows a rapid increase in the proportion of firms adopting AI, nearly reaching 100% before settling back to approximately 75%. This initial surge and subsequent decline highlight the impact of the data corruption channel: the reduced costs encourage widespread AI adoption, which in turn hurts the aggregate data quality, mitigating the advantages of adopting AI technology. The bottom left panel tracks the decline in aggregate data quality over time (γ_t), a direct consequence of increased reliance on AI, which, as illustrated in the bottom right panel, leads to a reduction in AI's relative productivity and, consequently, its attractiveness.[24] This experiment suggests that the labor displacement effects observed with the introduction of AI may partially reverse over time. This reversion effect could be significant, as demonstrated in this numerical example.

Figure 8 compares equilibrium functions under varying AI costs. The information law of motion when AI costs are high (represented by the yellow line) is located above that when the costs are low (indicated by the red line). Hence, if we begin with a steady state characterized by high costs of AI, following exogenous shocks that reduces the AI costs, the equilibrium information quality would gradually diminish over time and converge to the low steady state associated with lower costs of AI.

4.4 Equal Information Quality: A Counterfactual

To understand the importance of the data corruption channel,we conduct a counterfactual experiment where AI and human activities generate information of identical quality, setting $\gamma_A = \gamma_L$ to 100 for both, while other parameters remain as they were in the baseline calibration. Results are displayed in figure 9, comparing the baseline (blue solid line) against the scenario of equal information quality (red dashed line).

The top left panel shows that, in this counterfactual, the cost reduction needed to achieve 100% AI adoption in the short term is almost the same as in the baseline. However, the adoption dynamics differ markedly; with equal information quality, there is no significant reversal in AI adoption rates, which remain close to 100% in the long run, in stark contrast to the 30% reversal observed in

[24]The relative productivity of AI is calculated as the percentage differences in expected ex-post payoffs between AI adoption and labor adoption. Specifically it is equal to:

$$\frac{V^A(R_t) - E(V^l(\phi^i))}{E(V^l(\phi^i)}$$

where $V^A(R_t)$ and $V^l(\phi^i)$ are ex-post payoffs and are given by equation 18 and 17. The expectation is taken over idiosyncratic labor costs ϕ^i.

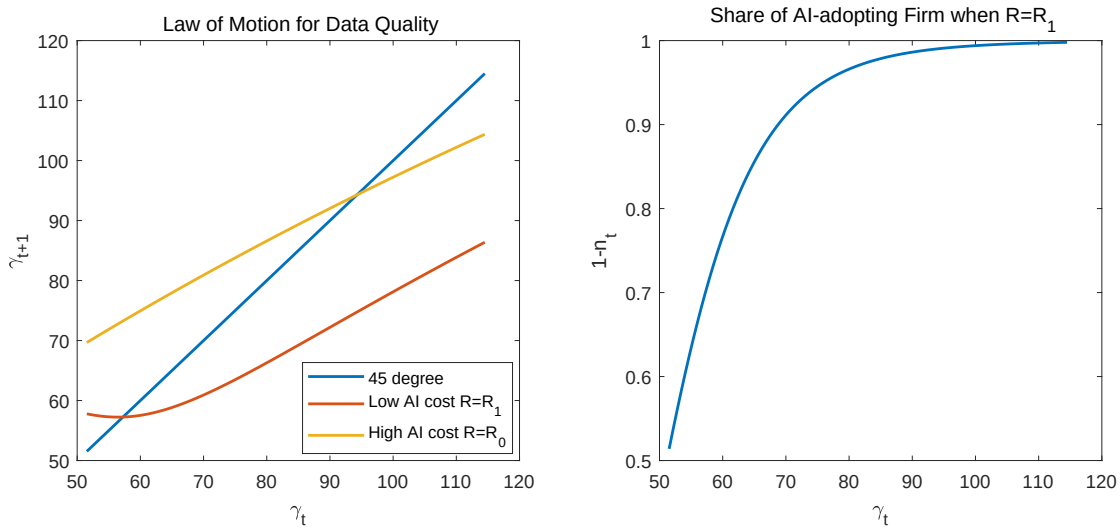

Figure 8: Equilibrium Functions when $\gamma_A = \gamma_L$

the baseline. This difference arises because data quality does not deteriorate as significantly when AI generates information of comparable quality to humans. Interestingly, even with equal information quality, we observe a gradual decline in data quality over time, attributed to AI's lesser reliance on private information compared to humans, and hence AI' action reveals less private information and contribute less to aggregate data quality compared to humans. Equal information quality implies a reduced data corruption effect, leading to a slower decline in AI's relative productivity and sustaining its appeal as an efficient choice for firms in the long term.

This counterfactual experiment concludes that the data corruption channel has a significant long-term impact on the labor market by diminishing AI's productivity. In our baseline scenario, this channel is responsible for reversing 30% of AI's short-term labor displacement effect. This reversal is entirely absent when information quality is equalized between humans and AI.

4.5 The Role of Public Information

In our baseline calibration, we set the precision of the public signal, γ_S, to zero, which naturally raises the question of how the model's dynamics might be affected by lower aggregate uncertainty, indicated by a higher value of γ_S. This section explores changes in the model's behavior with a more precise public signal.

To start, I set the precision of the public signal γ_S to 1. With this change, other parameters need to be recalibrated, particularly the relative quality of information generated by AI versus human. Figure 10 illustrates why a recalibration is necessary by comparing two circumstances. The red

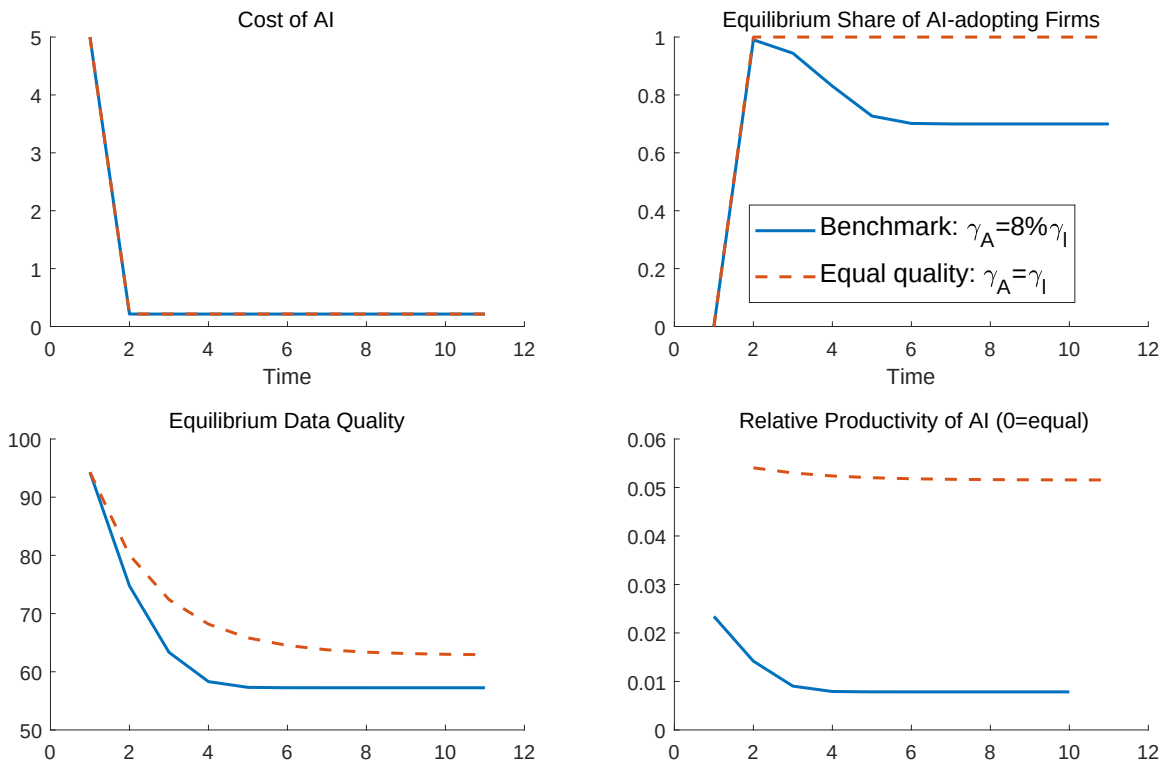

Figure 9: The ChatGPT Experiment with Equal Information Quality

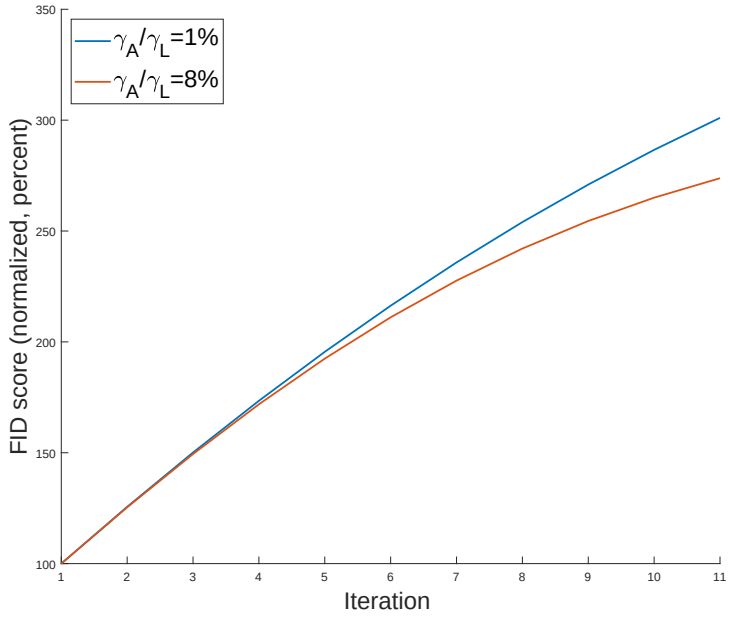

Figure 10: FID score with public signal $\gamma_S = 1$

line represents the case where the relative information quality remains at the benchmark level of 8%. However, with a more accurate public signal, this 8% difference is insufficient to achieve a 300% increase in the FID score. The lower uncertainty environment means that the contributions of human and AI to data quality are closer, as both are following the precise public signal, diminishing the data corruption effect and lessening the deterioration in data quality during full synthetic loops.

To replicate the observed 300% increase in the FID score, we find that the gap in private information between AI and humans must be widened. A recalibration indicates that this gap corresponds to AI's relative information quality being at 1% (as shown by the blue line in Figure 10). All other parameters remain the same as the benchmark calibration.

Figure 11 presents the numerical results when we conduct the ChatGPT experiment under this scenario of low uncertainty. The blue line is the benchmark result for comparison purposes. The red dashed line is the case with low uncertainty. Following the shock that reduces AI adoption costs, the immediate spike in the share of AI-adopting firms mirrors the original experiment, almost reaching 100%. Intriguingly, the reversal effect observed is quite similar to that of the benchmark scenario (as depicted in the top right panel of Figure 11), with the equilibrium share of AI-adopting firms stabilizing around 75%. This suggests that even under conditions of low uncer-

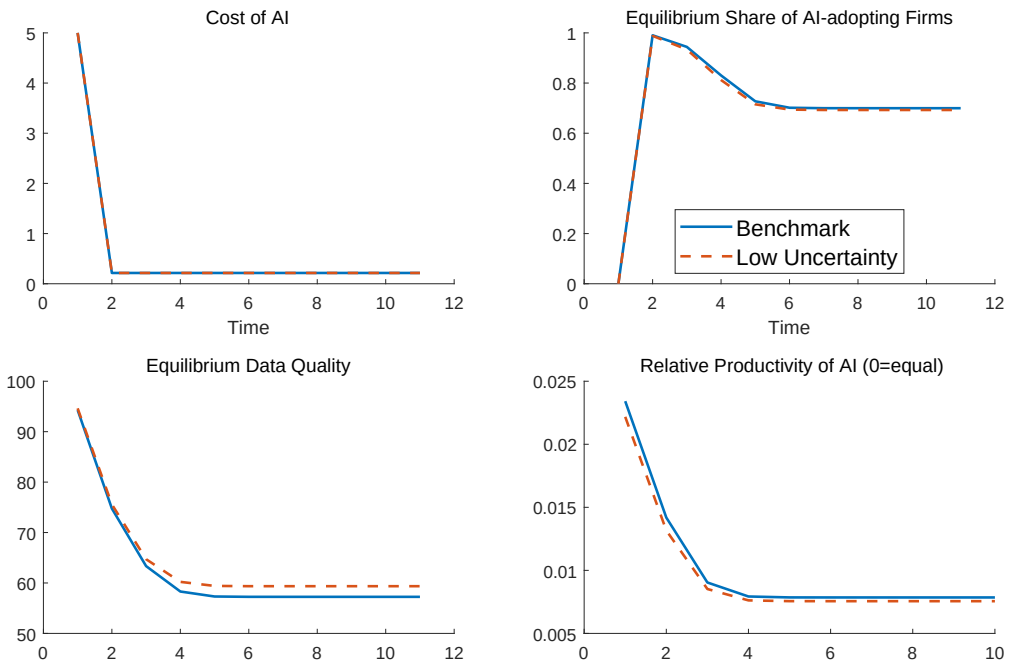

Figure 11: Model Dynamics with Low Uncertainty

tainty, the data corruption channel remains significant enough to counteract 30% of the short-term labor displacement effect.

We conclude that varying the precision of the public signal does not fundamentally change the impact of the data corruption channel. More critically, it underscores that when the model is calibrated to replicate the increase in FID score observed during synthetic loops as in the AI literature, the quantitative magnitude of the data corruption channel is effectively pinned down, rendering it robust to changes in other model parameters.

4.6 Review of Evidence from Computer Science

This section examines evidence in support of the model's key mechanism. The primary proposition of the model, which results in less labor displacement over the long run, is that increased AI adoption tends to produce less useful data points, subsequently reducing productivity for future AI. There is substantial evidence in computer science to support it. A recent paper reveals that the effectiveness of ChatGPT has decreased over time (Chen et al. 2023). The authors evaluate the performance of the March version of ChatGPT against its June version across a range of tasks such as prime-finding and code generation. They conclude that in most of these tasks, ChatGPT's

capability to generate correct responses has lessened.

What causes this decline? A major factor of concern is that since the introduction of ChatGPT, and large language models (LLMs) in general, they have been broadly adopted to perform a variety of tasks including text and image generation, app development, and more, thereby generating vast amounts of content posted on the internet. This AI-generated content is then utilized to train successive generations of AI models, which might have unforeseen consequences (Martínez et al. 2023). This aligns precisely with the feedback mechanism proposed in the paper.

Concurrent with this paper, contemporary scientific research in computer science has expressed similar worries, suggesting that the use of model-generated data to train an LLM could result in potentially problematic outcomes, such as quality degradation, reinforced biases, and loss of novelty. Alemohammad et al. (2023) examines generative image models trained on model-generated data and finds that training these models on synthetic data, as opposed to human-generated data, tends to progressively amplify artifacts (refer to their Figure 1). Consequently, "...models are destined to see a progressive decline in their quality (precision) or diversity (recall)." Shumailov et al. (2024) employs large language models like ChatGPT and shows that the use of model-generated content leads to "irreversible defects in the resulting models." They show that this phenomenon can occur in a variety of AI models, not limited to large language models. Research across various contexts and employing different AI software consistently shows that training AI models with synthetic content tends to degrade the quality of outputs, particularly affecting their diversity, according to studies such as Guo et al. (2023) and Doshi and Hauser (2024). The extent of this decline in data quality is influenced by the mix of real versus synthetic data in the training set, as highlighted by Briesch et al. (2023).

Additionally, there is evidence of a displacement effect where generative AI crowds out the production of human-generated content. Rio-Chanona et al. (2023) utilizes a difference-in-difference approach to compare the United States with Russia and China, where access to ChatGPT is more restricted. Their findings suggest that the launch of ChatGPT is associated with a 16% decrease in weekly contributions to Stack Overflow, indicating a significant crowding-out effect of generative AI on human contributions.

A final caveat of our calibration analysis is that it is confined to tasks performed by generative AI, as the experimental evidence is conducted solely within this domain. It is possible that for other tasks, AI might contribute more significantly to overall data quality. For example, as demonstrated in the model dynamics section, AlphaGo Zero was able to generate high-quality

data through self-play, ultimately surpassing even the best human players in Go.

Our numerical results, therefore, should not be interpreted as universally applicable to all economic tasks. Instead, they highlight an approach for combining economic models with experimental evidence from computer science to better estimate and inform key, deep parameters of those models. We believe this interdisciplinary approach holds great promise for future research at the intersection of AI and economics.

5 Efficiency and Regulation of AI

The regulation of AI has been a major topic in the ongoing debate about its impact. For example, the EU's AI Act, the first law specifically for AI, places restrictions on how AI can be adopted and how data can be used in different areas. In this section, we study the efficiency property of the competitive information equilibrium, focusing on AI adoption decisions and its use of data.

We focus on two main questions:

1. Is the private sector's decision to adopt AI socially efficient? Would a benevolent social planner, if given the chance, make different decisions about how many companies should adopt AI?

2. If the social planner could limit AI's access to certain data, would it optimal to do so? Would restricting AI's access to data lead to welfare improvement?

5.1 Formulation of Planner's Problem

To explore these questions, we set up a constrained social planner's problem. The planner is constrained in the sense that he faces the same information friction as private individuals, meaning that they cannot observe θ_t and also cannot directly interfere with the data accumulation process. However, the planner can make decisions on two margins: the labor threshold $\bar{\varphi}$ which determines the share of AI adoption, and the data available to the AI:

$$\Omega_t^A \subseteq \Omega_t, \forall t$$

where Ω_t^A represents the subset of historical data that the AI is allowed to access. The AI's decision problem is thus:

$$W^A(S_t, s_t^{Ai}, \Omega_{t-1}^A) = \max_{a_t^i} E(A_t^i(a_t^i)|S_t, s_t^{Ai}, \Omega_{t-1}^A) \tag{22}$$

The following lemma highlights the potential tradeoff faced by the social planner when considering restricting AI's access to historical data:

Lemma 5.1 *Given any $\Omega^A_{t-1} \subseteq \Omega_{t-1}$, the optimal action of AI is given by*

$$a^i_t = E(\theta_t|\Omega^A_{t-1}, S_t, s^{Ai}_t) = \frac{\gamma_S}{\gamma_S + \gamma_A + \kappa\gamma_t} S_t + \frac{\kappa\gamma_t}{\gamma_S + \gamma_A + \kappa\gamma_t}\mu_t + \frac{\gamma_A}{\gamma_S + \gamma_A + \kappa\gamma_t} s^{Ai}_t. \tag{23}$$

And the value of AI production is given by:

$$V^A = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa\gamma_t} - R_t \tag{24}$$

where κ is the ratio of conditional precision given Ω_t and Ω^A_t:

$$\kappa = \frac{1/Var\{\theta_t|\Omega^A_{t-1}\}}{1/Var\{\theta_t|\Omega_{t-1}\}} \leq 1$$

where the last inequality is due to $\Omega^A_{t-1} \subseteq \Omega_{t-1}$.

The direct cost of limiting AI's access to data is that it lowers AI productivity, as seen in equation 24, where introducing $\kappa < 1$ reduces the value of V^A. What's important is not the specific data points AI can access but the overall quality of the dataset. Therefore, the planner's decision comes down to choosing the optimal value of κ.

From an individual perspective, there is no advantage to restricting AI's access to data because V^A increases with κ. However, from a societal standpoint, there is a potential benefit. Restricting AI's access to public data (lowering κ) forces the AI to rely more on its privately-generated information. This can be seen in equation 23, where the weight on AI's private data s^{Ai}_t, $\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma_t}$, increases as κ decreases. Limiting AI's access to public data "forces" it to focus more on its own private data, potentially enhancing the quality of the overall dataset, especially in an economy where AI plays a significant role:

Lemma 5.2 *Given the current data quality γ_t and some share of labor adoption n_t, the future data quality γ_{t+1} is given by:*

$$\frac{1}{\gamma_{t+1}} = \rho^2 \frac{1}{\gamma_t + \gamma_s + n_t \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma_t}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta}$$

Thus, reducing AI's access to data (lowering κ) leads to higher future data quality (increasing γ_{t+1}).

We now formulate the constrained social planner's problem at the steady state. The planner can choose the relative share of AI adoption, $\bar{\varphi}$, as well as the quality of data available to AI, κ:

$$\max_{\bar{\varphi},\kappa} \int^{\bar{\varphi}} V^l(\varphi)\, d\varphi + \int_{\bar{\varphi}} V^A d\varphi$$

$$s.t.$$

$$\frac{1}{\gamma} = \rho^2 \frac{1}{\gamma + \gamma_s + n * \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1-n) * \left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta}$$

$$n = F(\bar{\varphi})$$

$$V^l(\varphi) = \bar{A} - \frac{1}{\gamma_s + \gamma_L} - \varphi$$

$$V^A = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa\gamma} - R$$

$$\kappa \leq 1$$

The social planner's objective is to maximize the ex-ante utility of entrepreneurs. The planner can choose the threshold $\bar{\varphi}$, which determines the share of labor versus AI-adopting entrepreneurs (n). This, in turn, affects the steady-state data quality γ, as described in the first constraint. From Lemma 5.2, we also know that a lower value of κ can enhance data quality. Together, these conditions jointly define $\gamma(\bar{\varphi}, \kappa)$ as a function of both $\bar{\varphi}$ and κ.

By substituting $\gamma(\bar{\varphi}, \kappa)$ and the value functions for labor adoption $V^l(\varphi)$ and AI adoption $V^A(\gamma)$, the social planner's problem can be written as:

$$\max_{\bar{\varphi},\kappa\leq 1} \int^{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S + \gamma_L} - \varphi\right] d\varphi + \int_{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa\gamma(\bar{\varphi}, \kappa)} - R\right] d\varphi \tag{25}$$

The key distinction between the social planner's problem and the decisions made by private agents is that the social planner understands how AI adoption and data access affect information quality (represented by the red expression $\gamma(.)$), while private agents treat γ as fixed. This creates an information externality that the planner needs to correct: private individuals, when making decisions, do not consider the broader impact of their actions on the overall data quality.

5.2 AI Adoption

We now focus on the AI adoption aspect of the social planner's problem. The first-order condition that determines the socially efficient threshold $\bar{\varphi}^{sp}$ is given by:

$$\underbrace{\frac{1}{\gamma_S + \gamma_A + \kappa\gamma(\bar{\varphi}^{sp})} + R - \frac{1}{\gamma_S + \gamma_l} - \bar{\varphi}^{sp}}_{\text{Private Benefits}} + \underbrace{\frac{(1 - n(\bar{\varphi}^{sp}))}{(\gamma_S + \gamma_A + \kappa\gamma(\bar{\varphi}^{sp}))^2}\gamma'(\bar{\varphi}^{sp})}_{\text{Externality}} = 0$$

This optimality condition consists of two parts: the "private benefits" and the "externality." The "private benefits" term mirrors the condition for individual optimization, while the "externality" term represents what the social planner takes into account, but private agents do not. The externality term depends on how the data quality γ changes with the adoption threshold $\bar{\varphi}$.

The following theorem shows the direction of this derivative and hence the optimal regulation

by government:

Theorem 5.1 *1. Increasing AI adoption hurts data quality ($\gamma'(\bar{\varphi}^{sp}) > 0$) if and only if the relatively quality of information generated by AI is not too much higher than those generated by labor:*

$$\frac{\gamma_A}{\gamma_L} < \frac{\gamma_S + \gamma}{\gamma_S} \tag{26}$$

2. Given the presence of data corruption, AI adoption hurts data quality, hence it should be taxed.

Condition 26 is similar to condition 16, where the externality is tied to the relative quality of information generated by humans and AI. When human-generated information is of higher quality compared to AI, the externality of AI adoption is negative, meaning that more AI adoption lowers the overall data quality. This negative externality justifies the taxation of AI, even in the long run. Notably, if AI generates more precise data, but the difference in quality between γ_L and γ_A is not substantial, a tax should still be imposed. This is because AI is less efficient at incorporating new information into the aggregate data, and taxing AI reflects this relative inefficiency. Thus, based on our benchmark calibration, a tax on AI adoption is necessary to achieve socially optimal data quality.

A caveat of the theorem is that it focuses on the steady state, primarily considering long-term optimal government regulation. In the short term, the need for regulation might be less pressing due to the dynamic nature of information externalities: today's AI adoption decisions affect tomorrow's data quality, influencing future welfare. Therefore, the extent and nature of regulation would depend on the time horizon the government prioritizes and how much weight is given to future generations. For instance, if the government focuses solely on the welfare of the current generation, regulation may not seem as necessary.

Additionally, the theorem suggests that government regulation should vary by task, depending on the level of uncertainty and quality of existing historical data. For example, in tasks with minimal uncertainty, like Go-playing, the case for taxation is weaker. However, in areas with high uncertainty, such as forecasting market demand or developing new drugs, the theorem implies that more uncertainty calls for greater regulation and taxation. Thus, government policy should be tailored to specific tasks, with regulation determined on a case-by-case basis.

5.3 Data Access

We now turn our attention to the data access aspect of the planner's problem: determining the optimal choice of κ. From equation 25 the choice of κ affects overall welfare only through the

value of AI adoption. From individual AI's perspective, it is always optimal to use 100% of the data available:

$$V^A \Uparrow = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa \Uparrow \gamma} - R.$$

However, the social planner recognizes that the choice of κ also impacts aggregate data quality:

$$V^A = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa \Uparrow \gamma(\kappa) \Downarrow} - R$$

where $\gamma'(\kappa) < 0$ because constraining public data usage of AI forces it to act more aggressively upon its own privately generated data, raising overall data quality and hence the value of γ (see lemma 5.2).

Thus optimal κ solves:

$$\max_{\kappa} \kappa\gamma(\kappa)$$

which suggests that setting κ to less than 100% can potentially improve welfare. The first-order condition is straightforward:

$$\underbrace{\gamma}_{\text{Cost of Limiting Data Access}} + \underbrace{\kappa\frac{\partial\gamma}{\partial\kappa}}_{\text{Benefit of Limiting Data Access}} = 0$$

The first term, γ, represents the cost of restricting AI's access to data—the higher the quality of the available dataset, the greater the cost of limiting its use. The second term reflects the societal benefit of restricting AI's data access, as this can improve overall data quality and increase the productivity of future AI systems.

Theorem 5.2 *With data corruption, long-run data quality decreases. Given this reduction in quality, the cost of limiting AI's access to data becomes smaller, making such a policy more likely to be socially desirable.*

In the presence of data corruption, the quality of data deteriorates over time (see the bottom left panel of figure 7). As a result, the cost of restricting AI's access to data decreases. On the other hand, the benefit of limiting data access is less sensitive to changes in data quality. Therefore, as the value of γ declines, the net benefit of constraining AI's access to data increases, making it more likely that such restrictions can improve social welfare.

6 Extensions and Robustness

This section offers two extensions to the base model to examine its robustness. We first relax the assumption that entrepreneurs, upon birth, cannot observe any prior history. Instead, we assume

that they are born with the ability to observe a certain fraction of past information. We also relax the assumption that the private data generated by AI is unrelated to past history, assuming instead that the quality of the data may improve with the quality of historical data observed. In this expanded model, we outline the model equilibrium and discuss its implications.

In the baseline model, it's assumed that entrepreneurs are incapable of observing past data history, while AI can observe the entire history. This assumption is mainly for clarity and to underscore the informational differences between humans and machines, emphasizing that machines (AI) are more efficient at collecting and processing extensive historical big data. In this section, we relax this assumption and demonstrate that the primary findings of the paper remain unchanged.

We propose that each entrepreneur, upon their inception, is able to observe a fraction of past data, denoted by $\kappa < 1$. This can be interpreted as entrepreneurs being able to observe information within their local communities and networks, which only constitute a fraction of the total amount of information available in the entire economic system. For instance, an entrepreneur from Michigan may be very familiar with business practices, culture, and customer preferences in Michigan, but less familiar with those in Florida or New York. Conversely, AI can collect and process data from all states in the US and even from the rest of the world.

Specifically, assume that the information set an entrepreneur born with (denoted by superscript e) is given by:

$$\Omega_{t-1}^{e} = \{S_{t-j}, D_{t-j}^{i}\}_{j=1,2,3,\ldots,-\infty}^{i=1,2,3,\ldots\kappa\bar{N}}$$

Note that the superscript for the number of noises now spans from 1 to $\kappa\bar{N} < \bar{N}$. As we will take $\bar{N} \to \infty$, there will be no integer problem associated with this notation.

This information set admits a similar recursion as shown in Y:

$$\Omega_{t}^{e} = \Omega_{t-1}^{e} \cup \{S_t, X_t^{le}, X_t^{Ae}\}$$

where

$$X_t^{le} = \frac{1}{\kappa N_t} \sum_{i \leq \kappa N_t} \left(a_t^i + \varepsilon_t^{Di}\right)$$

$$X_t^{Ae} = \frac{1}{\kappa\left(\bar{N} - N_t\right)} \sum_{i > \kappa N_t}^{\kappa\bar{N}} \left(a_t^i + \varepsilon_t^{Di}\right)$$

Denote the summary statistics for human and AI in the evolution of information for entrepreneurs. Denote the prior information content to be $\gamma_t^e = \frac{1}{Var\left(\theta_t | \Omega_{t-1}^e\right)}$.

We also relax the assumption that the precision of the signal generated by AI is independent

of the quality of the historical data the AI has been trained on. We refine this by postulating that the precision of the AI-generated signal follows the following functional form:

$$\gamma_A(\gamma_t) = a + b\gamma_t \tag{27}$$

where a and $b \geq 0$ are some constants. When $b = 0$, we go back to our baseline assumption.

Now, let's derive the model dynamics. We'll begin by determining the optimal actions for entrepreneurs choosing different modes of production.

If the agent chooses to use labor, his optimal action is:

$$a_t^i = E(\theta_t|\Omega_{t-1}^e, S_t, s_t^{li}) = \frac{\gamma_S}{\gamma_S + \gamma_L + \gamma_t^e} S_t + \frac{\gamma_t^e}{\gamma_S + \gamma_L + \gamma_t^e} \mu_t^e + \frac{\gamma_L}{\gamma_S + \gamma_L + \gamma_t^e} s_t^{li}$$

where γ_t is given by equation 10 and $\mu_t^e = E\left(\theta_t|\Omega_{t-1}^e\right)$ is the conditional mean of θ_t given past information, and the only informative part in this action is $\frac{\gamma_L}{\gamma_S+\gamma_L+\gamma_t^e} s_t^{Ai}$. Thus, it can be shown that the summary statistics for labor is (taking $\bar{N} \to \infty$):

$$X_t^{le} \sim N\left(\theta_t, \frac{1}{\kappa n_t}\left(\frac{\gamma_S + \gamma_L + \gamma_t^e}{\gamma_L}\right)^2 \frac{1}{\gamma_D}\right)$$

We now derive the signal associated with AI activities. The optimal action of AI is given by:

$$a_t^i = E(\theta_t|\Omega_{t-1}, S_t, s_t^{Ai}) = \frac{\gamma_S}{\gamma_S + \gamma_A(\gamma_t) + \gamma_t} S_t + \frac{\gamma_t}{\gamma_S + \gamma_A(\gamma_t) + \gamma_t} \mu_t + \frac{\gamma_A(\gamma_t)}{\gamma_S + \gamma_A(\gamma_t) + \gamma_t} s_t^{Ai}$$

where $\gamma_A(\gamma_t)$ is given by equation 27. Note that the only information content here is $\frac{\gamma_A(\gamma_t)}{\gamma_S+\gamma_A(\gamma_t)+\gamma_t} s_t^{Ai}$.

Hence, the summary statistic for AI activity is (taking $\bar{N} \to \infty$):

$$X_t^{Ae} \to N(\theta_t, \frac{1}{\kappa(1-n_t)}\left(\frac{\gamma_S + \gamma_A(\gamma_t) + \gamma_t}{\gamma_A(\gamma_t)}\right)^2 \frac{1}{\gamma_D})$$

With these two summary signals, we can derive the law of motion for γ_t and γ_t^e in a joint manner. Note that we need to keep track of both measures of information as they enter into the law of motion for both variables. Specifically, γ_{t+1} is given by the following function of γ_t and γ_t^e:

$$\frac{1}{\gamma_{t+1}} = \rho^2 \frac{1}{\gamma_t + \gamma_s + n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L+\gamma_t^e}\right)^2 \gamma_D + (1-n_t)\left(\frac{\gamma_A(\gamma_t)}{\gamma_S+\gamma_A(\gamma_t)+\gamma_t}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta} \tag{28}$$

And γ_{t+1}^e is given by the following function of γ_t and γ_t^e:

$$\frac{1}{\gamma_{t+1}^e} = \rho^2 \frac{1}{\gamma_t^e + \gamma_s + \kappa n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L+\gamma_t^e}\right)^2 \gamma_D + \kappa(1-n_t)\left(\frac{\gamma_A(\gamma_t)}{\gamma_S+\gamma_A(\gamma_t)+\gamma_t}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta} \tag{29}$$

From the two law of motion, one can examine how changes in n_t affects knowledge accumulation by examining whether

$$\left(\frac{\gamma_L}{\gamma_S+\gamma_L+\gamma_t^e}\right)^2-\left(\frac{\gamma_A\left(\gamma_t\right)}{\gamma_S+\gamma_A\left(\gamma_t\right)+\gamma_t}\right)^2>0$$

which boils down to:

$$\frac{\gamma_L}{\gamma_A\left(\gamma_t\right)}>\frac{\gamma_S+\gamma_t^e}{\gamma_S+\gamma_t}$$

when this condition is satisfied, reducing n_t (or more AI adoption) reduces knowledge accumulation for both AI and entrepreneurs.

The insights obtained from the baseline model remain. First, given that AI has access to more data points than humans, $\gamma_t>\gamma_t^e$, and therefore,

$$\frac{\gamma_S+\gamma_t^e}{\gamma_S+\gamma_t}<1$$

Hence, even if human and AI generate signals of equal precision, we would still expect AI adoption to hinder knowledge accumulation. However, now an offsetting force is introduced: γ_A is increasing in γ_t. So when γ_t is very large, γ_A would also be large, which would make the inequality harder to satisfy. This is due to the fact that the signal generated by AI improves in quality in relation to the overall data quality. However, this offsetting force is likely not strong enough to overturn the main result of the paper, as we will see in the full model dynamics.

To close the model, we now derive the endogenous share of labor-adopting entrepreneurs. In this case the value of adopting labor is:

$$V^l=\bar{A}-\frac{1}{\gamma_S+\gamma_L+\gamma_t^e}-\varphi_t^i$$

while the value of adopting AI is

$$V^A=\bar{A}-\frac{1}{\gamma_S+\gamma_A\left(\gamma_t\right)+\gamma_t}-R_t$$

Hence the threshold cost of labor is pinned down by

$$\bar{\varphi}\left(\gamma_t,\gamma_t^e\right)=R_t+\frac{1}{\gamma_S+\gamma_A\left(\gamma_t\right)+\gamma_t}-\frac{1}{\gamma_S+\gamma_L+\gamma_t^e} \tag{30}$$

Thus model dynamics is fully characterized by equation 28 and 29, where the share of labor-adopting entrepreneurs is given by

$$n_t\left(\gamma_t,\gamma_t^e\right)=F\left(\bar{\varphi}\left(\gamma_t,\gamma_t^e\right)\right)$$

and $\bar{\varphi}\left(\gamma_t,\gamma_t^e\right)$ is given by equation 30.

In contrast to our benchmark model, this system is characterized by two endogenous state variables representing the information possessed by entrepreneurs and AI, respectively. Figure 12 simulates the economy and carries out the ChatGPT experiment similar to section 4. It is observed

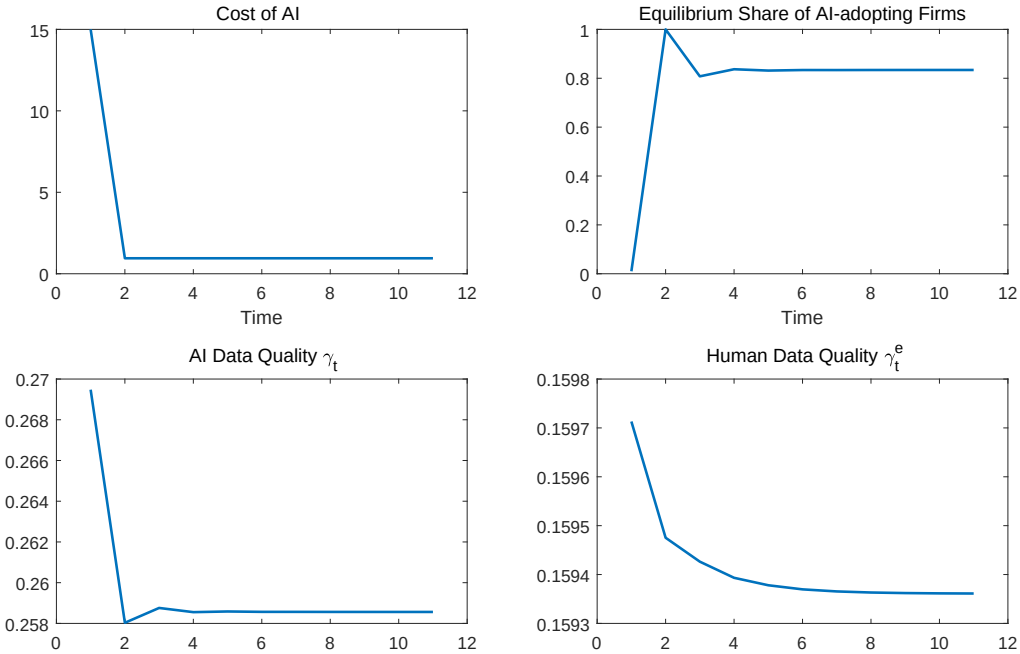

Figure 12: Model Dynamics with Low Uncertainty

that the labor displacement effect is also partially reversed in this case (top right panel). In this numerical example, I set $\kappa = 0.01$, indicating that the information possessed by humans is only 1% of that by AI. The difference in the overall level of information between γ_t and γ_t^e is much smaller, roughly equal to 60 percent (0.27 vs. 0.16) because of the existence of a public signal.[25]

Upon a reduction in the cost of AI, the share of AI-adopting firms increases. This displacement effect is partially reversed as in the baseline experiment. The logic for the reversal is basically the same as in the baseline experiment: the data quality of AI γ_t decreases (bottom left panel), reducing the efficacy and hence the productivity of AI products. However, there is an additional force here: the relative attractiveness of AI also depends on the efficiency of using labor, which in this case is endogenous to past information quality up to the coefficient κ. The bottom right panel indicates that the change in labor data quality γ_t^e decreases to a much lesser extent, indicating that there is virtually no change in the productivity of labor. This is because $\kappa < 1$, and therefore the productivity of labor is less affected by the endogenous evolution of data quality.

[25]Parameters used: $\rho = 0.95, \gamma_S = 0.05, \gamma_D = 2, \gamma_L = 0.1, \gamma_A = 0.1 + 0.1 * \gamma_t, \gamma_\eta = 0.5, \kappa = 0.01, R_t = 0.95, \forall t$. The function F follows a log normal distribution with a mean of -2 and a standard deviation of 2.

7 Conclusion

Data is the lifeblood of AI. The quality of data determines the quality of AI. With the recent rise of generative AI there has been a debate on the overall desirability of AI-generated data. This paper introduces a "data corruption" channel of AI-generated data and explores its implications for the long-term impact of AI-driven labor displacement. The paper presents a model where the adoption of AI and the quality of data are jointly determined in an information equilibrium. It shows that the desirabiliyt of AI-generated data depends on a key statistic: the relative quality of data produced by humans versus AI. When calibrated with recent findings from the computer science literature, the model suggests that AI-generated data is significantly lower in quality compared to human-generated data. As a result, the data corruption channel becomes quantitatively important, potentially reversing around 30% of the labor displacement effect over the long run. Despite this, AI regulation—such as through taxation—remains necessary in the long term to ensure socially optimal data quality.

The model is kept simple to emphasize the primary mechanism at play, but it can be extended to answer other important questions. For instance, while this paper assumes short-lived entrepreneurs for simplicity, future research could explore social learning in the context of long-lived firms. This would introduce additional complexities, such as data privacy issues highlighted in works like Jones and Tonetti (2020). A future exciting area of research would be to create a comprehensive quantitative model that considers various forces shaping the co-evolution of data and AI, allowing for a more detailed and practical discussion of government policies. Lastly, the interdisciplinary approach introduced in this paper could be applied to study economic tasks beyond those specifically addressed by generative AI, as studied in this paper.

References

Abis, Simona and Laura Veldkamp, "The Changing Economics of Knowledge Production," *Working Paper*, 2021.

Acemoglu, Daron, "The Simple Macroeconomics of AI," May 2024, (32487).

_ **and Pascual Restrepo**, "The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment," *American Economic Review*, June 2018, *108* (6), 1488–1542.

_ **and** _ , "Automation and New Tasks: How Technology Displaces and Reinstates Labor," *Journal of Economic Perspectives*, May 2019, *33* (2), 3–30.

_ **and** _ , "Tasks, Automation, and the Rise in U.S. Wage Inequality," *Econometrica*, 2022, *90* (5), 1973–2016.

_ **, David Autor, Jonathon Hazell, and Pascual Restrepo**, "Artificial Intelligence and Jobs: Evidence from Online Vacancies," *Journal of Labor Economics*, 2022, *40* (S1), S293–S340.

Aghion, Philippe, Benjamin F. Jones, and Charles I. Jones, *Artificial Intelligence and Economic Growth*, University of Chicago Press, January

Akerlof, George A., "The Market for "Lemons": Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics*, 1970, *84* (3), 488–500.

Alemohammad, Sina, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk, "Self-Consuming Generative Models Go MAD," 2023.

Alonso, Cristian, Andrew Berg, Siddharth Kothari, Chris Papageorgiou, and Sidra Rehman, "Will the AI revolution cause a great divergence?," *Journal of Monetary Economics*, 2022, *127*, 18–37.

Amador, Manuel and Pierre-Olivier Weill, "Learning from Prices: Public Communication and Welfare," *Journal of Political Economy*, 2010, *118* (5), 866–907.

Bohacek, Matyas and Hany Farid, "Nepotistically Trained Generative-AI Models Collapse," 2023.

Briesch, Martin, Dominik Sobania, and Franz Rothlauf, "Large Language Models Suffer From Their Own Output: An Analysis of the Self-Consuming Training Loop," 2023.

Brooks, Creston, Samuel Eggert, and Denis Peskoff, "The Rise of AI-Generated Content in Wikipedia," 2024.

Chen, Lingjiao, Matei Zaharia, and James Zou, "How is ChatGPT's behavior changing over time?," 2023.

Cong, Lin William, Danxia Xie, and Longtian Zhang, "Knowledge Accumulation, Privacy, and Growth in a Data Economy," *Management Science*, 2021, *67* (10), 6480–6492.

_ , **Wenshi Wei, Danxia Xie, and Longtian Zhang**, "Endogenous growth under multiple uses of data," *Journal of Economic Dynamics and Control*, 2022, *141*, 104395. Markets and Economies with Information Frictions.

Doshi, Anil R. and Oliver P. Hauser, "Generative AI enhances individual creativity but reduces the collective diversity of novel content," 2024.

Dowson, D.C and B.V Landau, "The FrÃ©chet distance between multivariate normal distributions," *Journal of Multivariate Analysis*, 1982, *12* (3), 450–455.

Fajgelbaum, Pablo D., Edouard Schaal, and Mathieu Taschereau-Dumouchel, "Uncertainty Traps*," *The Quarterly Journal of Economics*, 05 2017, *132* (4), 1641–1692.

Farboodi, Maryam and Laura Veldkamp, "A Model of the Data Economy," *Working Paper*, 2022.

_ , **Roxana Mihet, Thomas Philippon, and Laura Veldkamp**, "Big Data and Firm Dynamics," *AEA Papers and Proceedings*, May 2019, *109*, 38–42.

Guo, Yanzhu, Guokan Shang, Michalis Vazirgiannis, and Chloé Clavel, "The Curious Decline of Linguistic Diversity: Training Language Models on Synthetic Text," 2023.

Holcomb, Sean D., William K. Porter, Shaun V. Ault, Guifen Mao, and Jin Wang, "Overview on DeepMind and Its AlphaGo Zero AI," 2018, p. 67–71.

Jones, Charles I, "The A.I. Dilemma: Growth versus Existential Risk," Working Paper 31837, National Bureau of Economic Research November 2023.

Jones, Charles I. and Christopher Tonetti, "Nonrivalry and the Economics of Data," *American Economic Review*, September 2020, *110* (9), 2819–58.

Korinek, Anton and Donghyun Suh, "Scenarios for the Transition to AGI," March 2024, (32255).

Liu, Yang, Jiahuan Cao, Chongyu Liu, Kai Ding, and Lianwen Jin, "Datasets for Large Language Models: A Comprehensive Survey," 2024.

Martínez, Gonzalo, Lauren Watson, Pedro Reviriego, José Alberto Hernández, Marc Juarez, and Rik Sarkar, "Towards Understanding the Interplay of Generative Artificial Intelligence and the Internet," 2023.

Moll, Benjamin, Lukasz Rachel, and Pascual Restrepo, "Uneven Growth: Automation's Impact on Income and Wealth Inequality," *Econometrica*, 2022, *90* (6), 2645–2683.

Morris, Stephen and Hyun Song Shin, "Social Value of Public Information," *American Economic Review*, December 2002, *92* (5), 1521–1534.

Ordonez, Guillermo, "The Asymmetric Effects of Financial Frictions," *Journal of Political Economy*, 2013, *121* (5), 844 – 895.

Rio-Chanona, Maria Del, Nadzeya Laurentsyeva, and Johannes Wachs, "Are Large Language Models a Threat to Digital Public Goods? Evidence from Activity on Stack Overflow," 2023.

Shumailov, Ilia, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson, "AI models collapse when trained on recursively generated data," *Nature*, 2024, *631*, 755–759.

Silver, David, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy Lillicrap, Fan Hui, Laurent Sifre, George van den Driessche, Thore Graepel, and Demis Hassabis, "Mastering the game of Go without human knowledge," *Nature*, October 2017, *550*, 354–.

Trammell, Philip and Anton Korinek, "Economic Growth under Transformative AI," October 2023, (31815).

Veldkamp, Laura, "Slow boom, sudden crash," *Journal of Economic Theory*, 2005, *124* (2), 230–257.

Appendix

A Proof

Proof of Lemma 3.1

The optimal action a_t^i solves:

$$\max_{a_t^i} E\left[A_t^i\left(a_t^i\right)|S_t s_t^{Li}\right] = \max_{a_t^i} E\left[\bar{A} - \left(\theta_t - a_t^i\right)^2 |S_t, s_t^{Li}\right]$$

Take first order condition with respect to a_t^i :

$$a_t^i = E\left(\theta_t | S_t, s_t^{Li}\right)$$

Given that both S_t and s_t^{Li} are unbiased, normally-distributed signal of θ_t, we can apply the formula for conditional mean with normal distribution to obtain:

$$a_t^i = E\left(\theta_t | S_t, s_t^{Li}\right) = \frac{\gamma_S}{\gamma_S + \gamma_L} S_t + \frac{\gamma_L}{\gamma_S + \gamma_L} s_t^{Li}$$

a similar argument can be applied to derive the optmal action rule of AI.

Proof of Lemma 3.2

From the derivation we know that

$$\begin{aligned} X_t^l &= \frac{1}{N_t}\sum_{i=1}^{N_t}\left(\theta_t + \varepsilon_t^{Li} + \frac{\gamma_S + \gamma_L}{\gamma_L}\varepsilon_t^{Di}\right) \\ &= \theta_t + \frac{1}{N_t}\sum_{i=1}^{N_t}\varepsilon_t^{Li} + \frac{\gamma_S + \gamma_L}{\gamma_L}\frac{1}{N_t}\sum_{i=1}^{N_t}\varepsilon_t^{Di} \end{aligned}$$

Hence

$$\begin{aligned} Var\left(X_t^l|\theta_t\right) &= \frac{1}{N_t^2}N_t \underbrace{\frac{1}{\gamma_L}}_{\text{Variance of } \varepsilon_t^{Li}} + \left(\frac{\gamma_S + \gamma_L}{\gamma_L}\right)^2 \frac{1}{N_t^2}N_t \underbrace{\frac{\bar{N}}{\gamma_D}}_{\text{Variance of } \varepsilon_t^{Di}} \\ &= \frac{1}{N_t}\frac{1}{\gamma_L} + \left(\frac{\gamma_S + \gamma_L}{\gamma_L}\right)^2 \frac{\bar{N}}{N_t}\frac{1}{\gamma_D} \end{aligned}$$

When both $\bar{N}$ and $N_t \to 0$, the variance becomes:

$$Var\left(X_t^l|\theta_t\right) \to \left(\frac{\gamma_S + \gamma_L}{\gamma_L}\right)^2 \frac{1}{n_t \gamma_D}$$

where $n_t = \frac{N_t}{\bar{N}}$.

Hence the precision of the labor signal X_t^l is $n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D$.

Proof of Lemma 3.3

The AI signal data is given by

$$D_t^i = \frac{\gamma_S}{\gamma_S + \gamma_A + \gamma_t} S_t + \frac{\gamma_t}{\gamma_S + \gamma_A + \gamma_t}\mu_t + \frac{\gamma_A}{\gamma_S + \gamma_A + \gamma_t} S_t^{Ai} + \varepsilon_t^{Di}$$

Given that S_t and $\mu_t = E(\theta_t|\Omega_{t-1})$ are public information, the valuable piece in observating D_t^i is

$$\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}S_t^{Ai}+\varepsilon_t^{Di}$$

which is equivalent to

$$S_t^{Ai}+\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\varepsilon_t^{Di}$$

Hence the AI signal can be summarized as:

$$\begin{aligned} X_t^A &= \frac{1}{\bar{N}-N_t}\sum_{N_t+1}^{\bar{N}}\left(\theta_t+\varepsilon_t^{Ai}+\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\varepsilon_t^{Di}\right) \\ &= \theta_t+\frac{1}{\bar{N}-N_t}\sum_{N_t+1}^{\bar{N}}\varepsilon_t^{Ai}+\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\frac{1}{\bar{N}-N_t}\sum_{N_t+1}^{\bar{N}}\varepsilon_t^{Di} \end{aligned}$$

Hence the variance of the AI signal becomes:

$$\begin{aligned} & \frac{1}{(\bar{N}-N_t)^2}(\bar{N}-N_t)\frac{1}{\gamma_A}+\left(\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\right)^2\frac{1}{(\bar{N}-N_t)^2}(\bar{N}-N_t)\frac{\bar{N}}{\gamma_D} \\ = & \frac{1}{(\bar{N}-N_t)}\frac{1}{\gamma_A}+\left(\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\right)^2\frac{\bar{N}}{(\bar{N}-N_t)}\frac{1}{\gamma_D} \end{aligned}$$

When the number of enterepreneurs tends to infinity $\bar{N}$ and $N_t \to \infty$, the variance becomes

$$\left(\frac{\gamma_S+\gamma_A+\gamma_t}{\gamma_A}\right)^2\frac{1}{(1-n_t)\gamma_D}$$

where $n_t = \frac{N_t}{\bar{N}}$. hence the precision of the AI signal is

$$(1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2\gamma_D$$

.

Proff of Lemma 3.4

We start with the recursion:

$$\begin{aligned} Var(\theta_t|\Omega_t) &= Var\left(\theta_t|\Omega_{t-1},S_t,X_t^l,X_t^A\right) \\ &= \frac{1}{\gamma_t+\gamma_s+n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2\gamma_D} \end{aligned}$$

where the second equality follows from Lemma 3.2 and 3.3.

$$\frac{1}{\gamma_{t+1}} = Var(\theta_{t+1}|\Omega_t) = Var(\rho\theta_t+\eta_{t+1}|\Omega_t) = \rho^2 Var(\theta_t|\Omega_t)+\frac{1}{\gamma_\eta}$$

Plugging in the expression of $Var(\theta_t|\Omega_t)$, we obtained Lemma 3.4.

Proof of Theorem 3.1

To prove the theorem we need to evaluate the partial derivative of γ_{t+1} with respect to n_t :

$$\frac{\partial\gamma_{t+1}}{\partial n_t} \propto \left[\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2-\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2\right]\gamma_D$$

which is positive if and only if

$$\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2-\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\right)^2>0$$

Or

$$\frac{\gamma_L}{\gamma_S+\gamma_L}-\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}>0$$

Or

$$\gamma_L\left(\gamma_S+\gamma_A+\gamma_t\right)>\gamma_A(\gamma_S+\gamma_L)$$

Or

$$\gamma_L\gamma_S+\gamma_L\gamma_t>\gamma_A\gamma_S$$

$$\frac{\gamma_A}{\gamma_L}<\frac{\gamma_S+\gamma_t}{\gamma_S}$$

This is the condition 15.

Proof of Theorem 3.4

We take the labor signal quality to be fixed and the AI signa quality $\gamma_A\to\infty$, and therefore $\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma_t}\to 1$. The law of motion for aggregate data quality becomes:

$$\frac{1}{\gamma_{t+1}}=\rho^2\frac{1}{\gamma_t+\gamma_s+n_t\left(\gamma_t\right)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-n_t\left(\gamma_t\right)\right)\gamma_D}+\frac{1}{\gamma_\eta}$$

We assume that the labor cost distribution is compressed, so that small changes in the AI productivity would lead to large changes in the share of AI adoption.

We now construct multiple steady state as follows. Imagine a steady state where data quality is low, $\gamma=\gamma_1$, and so that everyone picks labor production:

$$n_t(\gamma_1)=1$$

Then γ_1 can be solved using the steady state equation:

$$\frac{1}{\gamma_1}=\rho^2\frac{1}{\gamma_1+\gamma_s+\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D}+\frac{1}{\gamma_\eta}$$

For another steady state with high data quality $\gamma=\gamma_2$, everyone chooses AI production:

$$n_t\left(\gamma_2\right)=0$$

Hence the equation to solve for γ_2 is:

$$\frac{1}{\gamma_2}=\rho^2\frac{1}{\gamma_2+\gamma_s+\gamma_D}+\frac{1}{\gamma_\eta}$$

Thus, to prove the existence of multiple steady states all we need to show is that

$$\gamma_1<\gamma_2$$

Then for some very compressed labor cost distribution we can have $n_t(\gamma_1)=1$ and $n_t\left(\gamma_2\right)=0$

given that $n(.)$ is decreasing in γ_t. This confirms the existence of multiple steady states.

To show that it suffices to show that the reasonable root of the following function about x is increasing in coefficient $m > 0$:

$$\frac{1}{x} = \rho^2 \frac{1}{x+m} + \frac{1}{\gamma_\eta} \tag{31}$$

If this can be proved, then given that $\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 < 1$, and thus $\gamma_s + \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D < \gamma_s + \gamma_D$, we can prove the claim that $\gamma_1 < \gamma_2$.

Note that equation 31 can be arranged as the following quadratic equation:

$$x + m = \rho^2 x + \frac{1}{\gamma_\eta} x (x+m)$$

$$\frac{1}{\gamma_\eta} x^2 + \left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) x - m = 0$$

and because $\frac{1}{\gamma_\eta} > 0$ and $m > 0$, this quadratic equation has a negative and a positive root, so the valid root is the positive one, or the larger root:

$$x = \frac{-\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + \sqrt{\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}}}{2\frac{1}{\gamma_\eta}}$$

We need to show that x is increasing in m.

Take the derivative:

$$\begin{aligned} x'(m) &= \frac{-\frac{1}{\gamma_\eta} + \frac{1}{2}\left[\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}\right]^{-\frac{1}{2}} \left(2\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)\frac{1}{\gamma_\eta} + 4\frac{1}{\gamma_\eta}\right)}{2\frac{1}{\gamma_\eta}} \\ &= \frac{-1 + \left[\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}\right]^{-\frac{1}{2}} \left(\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 2\right)}{2} \end{aligned}$$

Thus it suffices to show that

$$-1 + \left[\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}\right]^{-\frac{1}{2}} \left(\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 2\right) > 0$$

Or

$$\left[\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}\right]^{-\frac{1}{2}} \left(\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 2\right) > 1$$

Or

$$\left(\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 2\right) > \left[\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}\right]^{\frac{1}{2}}$$

Or

$$\left(\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 2\right)^2 > \left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}$$

Or

$$\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right) + 4 > \left(\rho^2 - 1 + \frac{m}{\gamma_\eta}\right)^2 + 4\frac{m}{\gamma_\eta}$$

Or

$$\rho^2 - 1 + \frac{m}{\gamma_\eta} + 1 > \frac{m}{\gamma_\eta}$$

Or

$$\rho^2 > 0$$

which is true by assumption. Hence we have proved the claim and the existence of multiple steady states is established.

Proof of Lemma 5.1

Denote $\gamma' = \frac{1}{Var\left(\theta_t|\Omega^A_{t-1}\right)}$. Hence we have

$$\kappa = \frac{\gamma'}{\gamma_t}$$
$$\gamma' = \kappa\gamma_t$$

Then the optimal action by AI is:

$$\begin{aligned} a^i_t &= E\left(a^i_t|\Omega^A_{t-1}, S_t, s^{Ai}_t\right) \\ &= \frac{\gamma_S}{\gamma_S + \gamma_A + \gamma'}S_t + \frac{\gamma'}{\gamma_S + \gamma_A + \gamma_t}\mu_t + \frac{\gamma_A}{\gamma_S + \gamma_A + \gamma'}S^{Ai}_t \\ &= \frac{\gamma_S}{\gamma_S + \gamma_A + \kappa\gamma_t}S_t + \frac{\kappa\gamma_t}{\gamma_S + \gamma_A + \gamma_t}\mu_t + \frac{\gamma_A}{\gamma_S + \gamma_A + \kappa\gamma_t}S^{Ai}_t \end{aligned}$$

Similarly we can easily derive:

$$V^A = \bar{A} - \frac{1}{\gamma_s + \gamma_A + \kappa\gamma_t} - R_t$$

Proof of Lemma 5.2

Define data quality in the same way as before:

$$\gamma_t = \frac{1}{Var\left(\theta_t|\Omega_{t-1}\right)}$$

Following a similar process as in the Lemma 3.3, one can show that X^A_t is an unbiased signal of θ_t with precision

$$(1 - n_t)\left(\frac{\gamma_A}{\gamma_S + \gamma_A + \kappa\gamma_t}\right)^2 \gamma_D$$

where $\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma_t}$ is the loading of AI's action on its privately generated data (see Lemma 5.1). Thus

$$\begin{aligned}
\frac{1}{\gamma_{t+1}} &= Var\left(\theta_{t+1}|\Omega_t\right) \\
&= \rho^2 Var\left(\theta_t|\Omega_t\right) + \frac{1}{\gamma_\eta} \\
&= \rho^2 Var\left(\theta_t|\Omega_{t-1}, S_t, X_t^l, X_t^A\right) + \frac{1}{\gamma_\eta} \\
&= \rho^2 \frac{1}{\gamma_t + \gamma_s + n_t\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D + (1-n_t)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma_t}\right)^2\gamma_D} + \frac{1}{\gamma_\eta}
\end{aligned}$$

Given that κ only shows up in the precision of the AI signal X_t^A, It is straightforward to see that γ_{t+1} increases when κ decreases, holding fixed other equilibrium quantities.

Proof of Theorem 5.1

Plug in the conditional value functions into the social planner's problem:

$$V^{sp} = \max_{\gamma,\bar{\varphi}} \int^{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S+\gamma_L} - \varphi\right] d\varphi + \int_{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S+\gamma_A+\gamma} - R\right] d\varphi$$

We could arrange the constraint as $\gamma\left(\bar{\varphi}\right)$, hence the value function becomes:

$$V^{sp} = \max_{\bar{\varphi}} \int^{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S+\gamma_L} - \varphi\right] d\varphi + \int_{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)} - R\right] d\varphi$$

Now consider perturbing $\bar{\varphi}$. The first order condition becomes:

$$\left[\bar{A} - \frac{1}{\gamma_S+\gamma_L} - \bar{\varphi}\right] - \left[\bar{A} - \frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)} - R\right] + \int_{\bar{\varphi}} \left[\frac{\gamma'\left(\bar{\varphi}\right)}{\left(\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)\right)^2}\right] d\varphi = 0$$

Thus the socially efficient threshold $\bar{\varphi}^{sp}$ is pinned down by:

$$\underbrace{\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}^{sp}\right)} + R - \frac{1}{\gamma_S+\gamma_L} - \bar{\varphi}^{sp}}_{\text{Private Benefit Term}} + \underbrace{\frac{\left(1-n\left(\bar{\varphi}^{sp}\right)\right)}{\left(\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}^{sp}\right)\right)^2}\gamma'\left(\bar{\varphi}^{sp}\right)}_{\text{Externality Term}} = 0$$

Compared to the private equilibrium where the threshold is pinned down by equating the private gains and benefits:

$$\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)} + R - \frac{1}{\gamma_S+\gamma_L} - \bar{\varphi} = 0$$

We have an extra term $\int_{\bar{\varphi}} \left[\frac{\gamma'(\bar{\varphi})}{(\gamma_S+\gamma(\bar{\varphi}))^2}\right] d\varphi$ which captures the positive externality: the fact that one's economic behavior could generate data, improve the data quality and the quality of AI, and therefore the overall welfare. Because of this extra term, the government needs to tax the AI in the steady state. Let's say that the government needs to impose a wedge (tax) τ on the return of AI, and rebate all proceeds in a lump-sum fashion to all the households in a uniform way. Then the

private CE becomes:

$$\left[\bar{A}-\frac{1}{\gamma_S+\gamma_L}-\bar{\varphi}\right]-\left[\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}-R\right](1-\tau)=0$$

$$\left[\bar{A}-\frac{1}{\gamma_S+\gamma_L}-\bar{\varphi}\right]-\left[\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}-R\right]+\tau\left[\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}-R\right]=0$$

$$\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}+R-\frac{1}{\gamma_S+\gamma_L}-\bar{\varphi}+\tau\left[\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}-R\right]=0$$

Hence we need the tax rate to be such that:

$$\tau\left[\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)}-R\right]=\int_{\bar{\varphi}}\left[\frac{\gamma'\left(\bar{\varphi}\right)}{\left(\gamma_S+\gamma_A+\gamma\left(\bar{\varphi}\right)\right)^2}\right]d\varphi$$

$$\tau=\frac{\left(1-F\left(\bar{\varphi}\right)\right)\left[\frac{\gamma'(\bar{\varphi})}{\left(\gamma_S+\gamma_A+\gamma(\bar{\varphi})\right)^2}\right]}{\bar{A}-\frac{1}{\gamma_S+\gamma_A+\gamma(\bar{\varphi})}-R}$$

where we factor out $\frac{\gamma'(\bar{\varphi})}{\left(\gamma_S+\gamma_A+\gamma(\bar{\varphi})\right)^2}$. But note that it depends on every term in this expression is positive except $\gamma'\left(\bar{\varphi}\right)$. Let us determine the sign of it. Write the constraint of the social planner as:

$$H\left(\gamma,\bar{\varphi}\right)=\frac{1}{\gamma}-\rho^2\frac{1}{\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D}-\frac{1}{\gamma_\eta}=0$$

$$\frac{\partial H}{\partial\bar{\varphi}}=\rho^2\frac{\left[\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D-\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right]f\left(\bar{\varphi}\right)}{\left(\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right)^2}$$

$$\frac{\partial H}{\partial\gamma}=-\frac{1}{\gamma^2}-\rho^2\frac{-\left(1-2\left(1-F\left(\bar{\varphi}\right)\right)\frac{\gamma_A}{\left(\gamma_S+\gamma_A+\gamma\right)^2}\gamma_D\right)}{\left(\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_l}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right)^2}$$

Hence

$$\gamma'\left(\bar{\varphi}\right)=-\frac{\frac{\partial H}{\partial\bar{\varphi}}}{\frac{\partial H}{\partial\gamma}}$$

One can show that $\frac{\partial H}{\partial\gamma}<0$:

$$\begin{aligned}\frac{\partial H}{\partial\gamma} &= -\frac{1}{\gamma^2}-\rho^2\frac{-1}{\left(\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right)^2}\\ &\quad -\rho^2\frac{2\left(1-F\left(\bar{\varphi}\right)\right)\frac{\gamma_A}{\left(\gamma_S+\gamma_A+\gamma\right)^2}\gamma_D^A}{\left(\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right)^2}\\ &< -\rho^2\frac{2\left(1-F\left(\bar{\varphi}\right)\right)\frac{\gamma_A}{\left(\gamma_S+\gamma_A+\gamma\right)^2}\gamma_D}{\left(\gamma+\gamma_s+F\left(\bar{\varphi}\right)\left(\frac{\gamma_l}{\gamma_S+\gamma_L}\right)^2\gamma_D+\left(1-F\left(\bar{\varphi}\right)\right)\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2\gamma_D\right)^2}<0\end{aligned}$$

because

$$\frac{1}{\gamma^2} > \rho^2 \frac{1}{\left(\gamma + \gamma_s + F(\bar{\varphi}) \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1 - F(\bar{\varphi})) \left(\frac{\gamma_A}{\gamma_S+\gamma_A+\gamma}\right)^2 \gamma_D\right)^2}$$

Hence the sign of $\gamma'(\bar{\varphi})$ depends on $\frac{\partial H}{\partial \bar{\varphi}}$, which in turn depends on

$$\left(\frac{\gamma_L}{\gamma_S + \gamma_L}\right)^2 \gamma_D - \left(\frac{\gamma_A}{\gamma_S + \gamma_A + \gamma}\right)^2 \gamma_D$$

Hence if

$$\frac{\gamma_L}{\gamma_A} > \frac{\gamma_S}{\gamma_S + \gamma(\bar{\varphi})}$$

we can show that

$$\left(\frac{\gamma_L}{\gamma_S + \gamma_L}\right)^2 \gamma_D - \left(\frac{\gamma_A}{\gamma_S + \gamma_A + \gamma}\right)^2 \gamma_D > 0$$

, then

$$\gamma'(\bar{\varphi}) > 0$$

which means that

$$\tau > 0$$

On the other hand, if AI is more efficient in generating data than labor:

$$\frac{\gamma_L}{\gamma_A} < \frac{\gamma_S}{\gamma_S + \gamma(\bar{\varphi})}$$

Then one can show that

$$\gamma'(\bar{\varphi}) < 0$$

and hence

$$\tau < 0$$

Proof of Theorem 5.2

The tradeoff of κ is that reducing κ tends to reduce the value of AI adoption, but at the same time allow AI to act more aggresively upon its own private information, thus incorproate more information into aggregate knowledge, raising the value of γ. In other words:

$$V^A = \bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa \downarrow \gamma(\kappa \uparrow)} - R$$

Note that the law of motion for γ can be written compactly as $\gamma(\bar{\varphi}, \kappa)$, Plug it into the objective function, we have:

$$\max_{\gamma, \bar{\varphi}, \kappa} \int^{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_s + \gamma_L} - \varphi\right] d\varphi + \int_{\bar{\varphi}} \left[\bar{A} - \frac{1}{\gamma_S + \gamma_A + \kappa\gamma(\bar{\varphi}, \kappa)} - R\right] d\varphi$$

$$s.t.$$

$$\kappa \leq 1$$

In terms of the optimal choice of κ, we just need to solve the very simple function:

$$\max \kappa\gamma\left(\bar{\varphi}, \kappa\right)$$

Or

$$\underbrace{\gamma}_{\text{Cost of reducing } \kappa} + \underbrace{\kappa\frac{\partial\gamma}{\partial\kappa}}_{\text{Benefit of Reducing } \kappa} = 0$$

We need to derive some partial derivatives. Define the law of motion for γ as:

$$H(\gamma, \bar{\varphi}, \kappa) = \rho^2 \frac{1}{\gamma + \gamma_s + n * \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1-n) * \left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2 \gamma_D} + \frac{1}{\gamma_\eta} - \frac{1}{\gamma} = 0$$

$$\frac{\partial\gamma}{\partial\kappa} = -\frac{\frac{\partial H}{\partial\kappa}}{\frac{\partial H}{\partial\gamma}} = -\frac{\rho^2 \frac{1}{\left[\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right]^2}(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^4}2\left(\gamma_S+\gamma_A+\kappa\gamma\right)*\gamma*\gamma_D}{\rho^2\frac{-1}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma^2}}$$

Note that

$$\frac{\partial H}{\partial\kappa} = \rho^2 \frac{1}{\left[\gamma + \gamma_s + n * \left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2 \gamma_D + (1-n) * \left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2 \gamma_D\right]^2}(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^4}2\left(\gamma_S+\gamma_A+\kappa\gamma\right)$$

$$\frac{\partial H}{\partial\gamma} = \rho^2 \frac{-\left(1-2\left(1-n\right)\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\frac{\gamma_A}{(\gamma_S+\gamma_A+\kappa\gamma)^2}\kappa\gamma_D\right)}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma^2} > 0$$

Hence

$$\frac{\partial\gamma}{\partial\kappa} < 0$$

Thus the optimal condition is given by

$$\frac{\partial\gamma}{\partial\kappa} = -\frac{\gamma}{\kappa}$$

Let's simplify the derivative $\frac{\partial\gamma}{\partial\kappa}$ and see how it changes with γ :

$$\frac{\partial\gamma}{\partial\kappa} = -\frac{\rho^2\frac{2(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^3}\gamma*\gamma_D}{\left[\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right]^2}}{\rho^2\frac{-\left(1-2(1-n)\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\frac{\gamma_A}{(\gamma_S+\gamma_A+\kappa\gamma)^2}\kappa\gamma_D\right)}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma^2}}$$

$$= -\frac{\gamma}{\kappa}\frac{\kappa}{\gamma}\frac{\rho^2\frac{2(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^3}\gamma*\gamma_D}{\left[\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right]^2}}{\rho^2\frac{-\left(1-2(1-n)\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\frac{\gamma_A}{(\gamma_S+\gamma_A+\kappa\gamma)^2}\kappa\gamma_D\right)}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma^2}}$$

$$= -\frac{\gamma}{\kappa}\frac{\rho^2\frac{2(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^3}\gamma*\gamma_D*\kappa}{\left[\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right]^2}}{\rho^2\frac{-\left(\gamma-2(1-n)\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\frac{\gamma_A}{(\gamma_S+\gamma_A+\kappa\gamma)^2}\kappa\gamma\gamma_D\right)}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma}}$$

The denominator becomes:

$$\rho^2\frac{-\left(\gamma-2\left(1-n\right)\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\frac{\gamma_A}{(\gamma_S+\gamma_A+\kappa\gamma)^2}\kappa\gamma\gamma_D\right)}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma}$$

$$= \quad \rho^2\frac{-\gamma}{\left(\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right)^2}+\frac{1}{\gamma}+N$$

where N is the numerator: $N=\rho^2\frac{2(1-n)*\gamma_A^2\frac{1}{(\gamma_S+\gamma_A+\kappa\gamma)^3}\gamma*\gamma_D*\kappa}{\left[\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D\right]^2}$

Note from the law of motion of γ :

$$\frac{1}{\gamma}=\rho^2\frac{1}{\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D}+\frac{1}{\gamma_\eta}$$

Thus

$$\frac{1}{\gamma+\gamma_s+n*\left(\frac{\gamma_L}{\gamma_S+\gamma_L}\right)^2\gamma_D+(1-n)*\left(\frac{\gamma_A}{\gamma_S+\gamma_A+\kappa\gamma}\right)^2\gamma_D}=\frac{\frac{1}{\gamma}-\frac{1}{\gamma_\eta}}{\rho^2}$$

Substitute:

$$-\rho^2\gamma\left(\frac{\frac{1}{\gamma}-\frac{1}{\gamma_\eta}}{\rho^2}\right)^2+\frac{1}{\gamma}$$

Hence the derivative becomes:

$$\frac{\partial\gamma}{\partial\kappa}=-\frac{\gamma}{\kappa}\frac{N}{\frac{1}{\gamma}-\rho^2\gamma\left(\frac{\frac{1}{\gamma}-\frac{1}{\gamma_\eta}}{\rho^2}\right)^2+N}$$

Now, substitute in the first order condition:

$$-\frac{\gamma}{\kappa}\frac{N}{\frac{1}{\gamma}-\rho^2\gamma\left(\frac{\frac{1}{\gamma}-\frac{1}{\gamma_\eta}}{\rho^2}\right)^2+N}=-\frac{\gamma}{\kappa}$$

We have the following equation to pin down γ :

$$\frac{1}{\gamma} - \rho^2 \gamma \left(\frac{\frac{1}{\gamma} - \frac{1}{\gamma_\eta}}{\rho^2} \right)^2 = 0$$

Hence this equation pins down the optimal γ when κ is endogenous. The optimum can be solved in a quadratic equation

$$\left(1-\rho^2\right) \left(\frac{1}{\gamma}\right)^2 - 2\frac{1}{\gamma_\eta}\frac{1}{\gamma} + \left(\frac{1}{\gamma_\eta}\right)^2 = 0$$

Hence

$$\begin{aligned} \left(\frac{1}{\gamma}\right)^* &= \frac{\frac{2}{\gamma_\eta} \pm \sqrt{\frac{4}{\gamma_\eta^2} - 4\left(1-\rho^2\right)\left(\frac{1}{\gamma_\eta}\right)^2}}{2\left(1-\rho^2\right)} \\ &= \frac{\frac{2}{\gamma_\eta} \pm \frac{2}{\gamma_\eta}\rho}{2\left(1-\rho^2\right)} = \frac{1}{\gamma_\eta}\frac{1}{1 \pm \rho} \end{aligned}$$

The root should be

$$\left(\frac{1}{\gamma}\right)^* = \frac{1}{\gamma_\eta}\frac{1}{1-\rho}$$

because

$$\frac{1}{\gamma} > \frac{1}{\gamma_\eta}$$

Hence to maintain optimal level of γ^*, with data corruption, the value of κ needs to go down. Hence

$$\gamma^* = \gamma_\eta \left(1-\rho\right)$$

and we can check the conditions under which this is satisfied or not.

We start with a steady state where the cost of AI adoption is very high. Hence the share of AI adoption is minimal ($n \approx 0$). In this steady state, the data quality is relatively high. Denote it by γ_h. Assume that $\gamma_h > \gamma^*$. To reduce the value of γ, we need to increase κ, but the maximum value of κ is 1. Hence at this steady state $\kappa = 1$.

We now examine a steady state with data corruption. From the numerical simulation we know that with data corruption, the AI adoption rate is positive: $\bar{\varphi} < \infty$ and $n < 100\%$. If the data corruption channel is sufficiently strong, without data regulation the value of steady state γ would drop by a lot. If $\gamma < \gamma^*$ as defined by previous equation, we need to set the value of κ to increase the value of γ. From the law of motion of γ and Lemma 5.2, we know that:

$$\frac{\partial \gamma}{\partial \kappa} < 0$$

Hence to raise the value of γ we need κ to decrease, which implies that the social planner potentially needs to constrain AI's access to data to achieve the desirable level of γ.

Table 1: Works on Data Corruption and AI Model Collapse

Title	*Authors*	*Date*
The Curse of Recursion: Training on Generated Data Makes Models Forget	Shumailov et al. (Oxford)	May 2023
Towards Understanding the Interplay of Generative Artificial Intelligence and the Internet	Martínez et al. (Madrid)	June 2023
Self Consuming Generative Model Go Mad	Alemohammad et al. (Stanford)	July 2023
Are Large Language Models a Threat to Digital Public Goods? Evidence from Activity on Stack Overflow	Rio-Chanona et al. (Harvard)	July 2023
Will Large-scale Generative Models Corrupt Future Datasets?	Hataya et al. (Kyoto)	Aug 2023
Generative artificial intelligence enhances individual creativity but reduces the collective diversity of novel content	Doshi et al. (UCL)	Aug 2023
The Curious Decline of Linguistic Diversity: Training Language Models on Synthetic Text	Guo et al. (École Polytechnique)	Nov 2023
Large Language Models Suffer From Their Own Output: An Analysis of the Self-Consuming Training Loop	Briesch et al. (Johannes Gutenberg)	Nov 2023
AI-Generated Images Introduce Invisible Relevance Bias to Text-Image Retrieval	Xu et al. (Chinese Academy of Sciences)	Nov 2023
Nepotistically Trained Generative-AI Models Collapse	Bohacek and Farid (Stanford)	Nov 2023
Under the Surface: Tracking the Artifactuality of LLM-Generated Data	Das et al. (U of Minnesota)	Jan 2024
Model Collapse Demystified: The Case of Regression	Dohmatob et al. (New York University)	Apr 2024
Self-Consuming Generative Models with Curated Data Provably Optimize Human Preferences	Ferbach et al. (Montréal)	June 2024
Towards Theoretical Understandings of Self-Consuming Generative Models	Fu et al. (USTC)	June 2024
Regurgitative Training: The Value of Real Data in Training Large Language Models	Zhang et al. (Tsinghua University)	July 2024

Note: Papers are listed in chronological order. For the sake of space, only the first academic affiliation is listed. The list does not meant to be exhaustive.